Foundations of Statistical Learning

DS5616 Lecture Notes

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"Learning is not about collecting nuggets and bringing them home;
It's about gnawing at our walls and bringing them down,
To make the whole world our home."

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Sections marked with $*$ are optional.

Notation

Notation	Meaning
$[n]$	$\{1, 2, \ldots, n\}$
2^{Ω}	set of all subsets of Ω
$X \perp\!\!\!\perp Y$	X and Y are independent
$X \sim P$	X has distribution P
$a \approx b$	a is approximately b
$a := b$	a is defined to be b
X^n	$X_1, \ldots, X_n$
$\ln n$	$\log_e n$
$\exp\{x\}$	e^x
$\|a\|_2 : a \in \mathbb{R}^n$	ℓ_2 norm of a: $\sqrt{a_1^2 + \cdots + a_n^2}$
WLOG	without loss of generality
w.r.t.	with respect to
iff	if and only if
w.p.	with probability
w.h.p.	with high probability

Chapter 1

Probability Essentials

We are interested in *random experiments* – experiments whose outcomes we cannot predict with certainty in advance. But when one repeats the experiment multiple times, one starts seeing some "regularity" in the average output. For instance, we cannot predict with certainty the outcome of a coin toss (assuming it is not a Sholay coin[1]). However, if we toss the coin a million times, we can be "fairly certain" that we will get an average of "about" 50% Heads (for a fair coin).

Probability theory seeks to build the necessary mathematics to describe such phenomena. It has three main ingredients:

1. *Sample space* – set of all possible outcomes of the experiment; usually denoted Ω.

   Examples:

   - A toss of a coin - $\Omega = \{H, T\}$
   - Two tosses of a coin - $\Omega = \{HH, HT, TH, TT\}$
   - A throw of two dice - $\Omega = \{(i, j) : 1 \leq i \leq 6, 1 \leq j \leq 6\}$

2. *Event space* – An "event" is a property that can be observed either to hold or not to hold after the experiment is done. In mathematical terms, an event is a subset of Ω. The event space, denoted $\mathcal{F}$, is a collection of subsets of Ω satisfying certain properties. Whenever $|\Omega|$ is finite, we let $\mathcal{F}$ to comprise all the subsets of Ω, denoted 2^{Ω}.

   Example: A toss of a coin - $\mathcal{F} = \{\phi, \{H\}, \{T\}, \{H, T\}\}$.

   The event $A \cap B$ is interpreted as event A and event B. Similarly, $A \cup B$ is A or B.

3. *Probability* – With each event A one associates a number between 0 and 1, denoted by $P(A)$, called the "probability of A". This number measures the likelihood of the event A to be realized *a priori*, before performing the experiment. The more likely the event is, the closer to 1 this number is. Formally, $P : \mathcal{F} \to [0, 1]$ should satisfy, for all $E_n \in \mathcal{F}$,

   (i) $P(\Omega) = 1$.

   (ii) $P(\cup_n E_n) = \sum_n P(E_n)$ whenever $E_n \cap E_m = \phi$ for all $m \neq n$.

Together, they specify a **probability space** $(\Omega, \mathcal{F}, P)$.

[1] The Sholay coin is a coin from the Bollywood movie Sholay (1975) that has the same image on both sides. In the movie, the character Jai uses this coin to settle disputes with his friend Viru by tossing it.

Exercise 1.1: Properties of P

1. For $\Omega = \{a_1, \ldots, a_n\}$ and $\mathcal{F} = 2^\Omega$, let $P(\{a_i\}) = p_i$ such that $p_i \in [0,1]$ for all $i \in [n]$ and $\sum_{i=1}^n p_i = 1$. For any set $A \in \mathcal{F}$, we define $P(A) = \sum_{a \in A} P(\{a\})$. Verify that properties (i) and (ii) above are satisfied.

2. Let $A, B, E_n \in \mathcal{F}$. Verify the following using (i) and (ii):

   - $P(\phi) = 0$.
   - If $A \subseteq B, P(A) \leq P(B)$
   - $P(A^c) = 1 - P(A)$
   - $P(A \cup B) + P(A \cap B) = P(A) + P(B)$
   - $P(\cup_n E_n) \leq \sum_n P(E_n)$

3. Let $\Omega = \{\heartsuit, \diamondsuit, \spadesuit, \clubsuit\}$ and let $\mathcal{F} = \{\phi, \{\diamondsuit, \heartsuit\}, \{\spadesuit, \clubsuit\}, \Omega\}$. Assign probabilities to the elements of $\mathcal{F}$ such that (i) and (ii) are satisfied.

   - Can you compute $P(A \cup B), P(A \cap B)$, and $P(A^c)$ for all $A, B \in \mathcal{F}$?
   - Can you compute $P(A \cup B), P(A \cap B)$, and $P(A^c)$ for all $A, B \in 2^\Omega$?

1.1 Conditional probability and Independence

Definition 1.1. Two events A and B are **independent** if

$$P(A \cap B) = P(A) \cdot P(B).$$

Exercise 1.2: Independence

Show that: If A and B are independent, so are A and B^c, A^c and B, and A^c and B^c.

Definition 1.2. Let A, B be events with $P(B) > 0$. The **conditional probability of A given B** is

$$P(A|B) = \frac{P(A \cap B)}{P(B)}.$$

Theorem 1.1. *Suppose $P(B) > 0$. Then,*

1. *A and B are independent if and only if $P(A|B) = P(A)$.*
2. *The operation $A \mapsto P(A|B)$ from $\mathcal{F} \to [0,1]$ defines a new probability distribution on $\mathcal{F}$, called the "conditional probability distribution given B".*

Exercise 1.3: Conditional probability

Prove Theorem 1.1.
Hint for (2): Let $Q(A) = P(A|B)$. Show that Q satisfies (i) and (ii) in the definition of probability.

Definition 1.3. A collection of events $(E_n)_{n \geq 1}$ is called a **partition of Ω** if

- $E_n \in \mathcal{F}$ for each n,
- E_n are pairwise disjoint; i.e., $E_n \cap E_m = \phi$ for all $m \neq n$.
- $P(E_n) > 0$ for each n, and
- $\cup_n E_n = \Omega$.

Theorem 1.2. *Let $(E_n)_{n\geq 1}$ be a partition of Ω. Then, if $A \in \mathcal{F}$, we have*

$$P(A) = \sum_n P(A|E_n)P(E_n).$$

Proof. We know that

$$P(A) = P(A \cap \Omega).$$

Since $\cup_n E_n = \Omega$, we can write

$$P(A) = P(A \cap (\cup_n E_n)).$$

By "distributing $\cap$ across $\cup$", we get

$$P(A) = P(\cup_n (A \cap E_n)). \tag{1.1}$$

Now, since E_n are disjoint, so are $A \cap E_n$. Therefore, by (ii) in the definition of probability, we have

$$P(\cup_n (A \cap E_n)) = \sum_n P(A \cap E_n). \tag{1.2}$$

By the definition of conditional probability, we can write

$$\sum_n P(A \cap E_n) = \sum_n P(A|E_n) P(E_n). \tag{1.3}$$

The proof is complete by putting together (1.1), (1.2), and (1.3). □

Theorem 1.3 (Bayes' theorem)**.** *Let $(E_n)_{n\geq 1}$ be a partition of Ω, and suppose $P(A) > 0$. Then*

$$P(E_n|A) = \frac{P(A|E_n)P(E_n)}{\sum_m P(A|E_m)P(E_m)}.$$

Exercise 1.4: Bayes' theorem

1. Prove Theorem 1.3.
2. Let $0 < P(B) < 1$ and A be any event. Show that
$$P(A) = P(A|B)P(B) + P(A|B^c)P(B^c).$$
3. Let A, B be events with $P(A) > 0$. Show that
$$P(A \cap B|A \cup B) \leq P(A \cap B|A).$$
4. Samples are screened for common cold. Suppose the test has 99% accuracy, and that one in ten thousand people are common cold positive. The test has a 5% false positive rating, as well. Suppose the test screens you as positive. What is the probability you have common cold?
Hint: 99% refers to P(test is positive|you have common cold). You want to calculate P(you have common cold|test is positive).

1.2 Random variables and Expectation

Definition 1.4. A **random variable** X on a probability space $(\Omega, \mathcal{F}, P)$ is a real-valued function on Ω; that is, $X : \Omega \to \mathbb{R}$ such that $X^{-1}(B) \in \mathcal{F}$ for all "reasonable" $B \subseteq \mathbb{R}$. Here

$$X^{-1}(B) = \{\omega \in \Omega : X(\omega) \in B\}.$$

One may imagine X to be a mapping from the probability space $(\Omega, \mathcal{F}, P)$ to another probability space $(\mathbb{R}, \mathcal{B}, P_X)$ where $\mathcal{B}$ comprises all "reasonable" $B \subseteq \mathbb{R}$. We assign probabilities P_X to events $B \in \mathcal{B}$ as

$$P_X(B) = P(X^{-1}(B)).$$

Note that this is possible since P is defined for all elements of $\mathcal{F}$ and $X^{-1}(B) \in \mathcal{F}$ by definition.
Notation: Sometimes, we write $\mathbb{P}[X \in B]$ to denote $P_X(B)$.

Further, (by properties of $\mathcal{B}$), specifying this probability P_X for sets of the form $(-\infty, x]$ suffices to specify P_X for all $B \in \mathcal{B}$. Towards this, we define a function $F_X(x)$ such that

$$F_X(x) = \mathbb{P}(X \leq x) = P_X((-\infty, x]) = P(A_x)$$

where

$$A_x = X^{-1}((-\infty, x]) = \{\omega \in \Omega : X(\omega) \leq x\}.$$

$F_X(\cdot)$ is called the cumulative distribution function (CDF) of X. The joint CDF of random variables X and Y is given by

$$F_{XY}(x, y) := \mathbb{P}(X \leq x, Y \leq y).$$

Definition 1.5. A **discrete random variable** is a random variable that takes no more than countably infinite many values. The vector comprising the probabilities of the values is called the **probability mass function** (PMF) of the random variable.

Definition 1.6. A **continuous random variable** is a random variable whose CDF is a continuous function. When the CDF F_X admits a density, i.e., there exists a function f_X such that

$$F_X(x) = \int_{-\infty}^{x} f_X(t)dt,$$

f_X is called the **probability density function** (PDF) of X.

For a discrete random variable, to define P_X, it suffices to specify the probabilities $P_X(\{x\})$ for all $x \in \mathbb{N}$.
Notation: It is customary to denote $P_X(\{x\})$ by $P_X(x)$; also, we assume that the unspecified probabilities $P_X(x)$ are zero for $x \in \mathbb{N}$. Sometimes, we use calligraphic letters to specify the "alphabet" of the random variable. For example, $X \in \mathcal{X}$ means $P_X(x) = 0$ for $x \notin \mathcal{X}$.

Exercise 1.5: CDF examples

Sketch the CDF $F_X(x)$ as a function of $x \in \mathbb{R}$, for

1. $P_X(1) = p = 1 - P_X(0)$.
2. $P_X(x) = 1/6$ for all $x \in \{1, 2, 3, 4, 5, 6\}$.
3. $P_X(k) = (1-p)^{k-1}p$ for $k = \{1, 2, \ldots\}$, $p \in (0, 1)$.

Two discrete random variables X and Y are independent if

$$\mathbb{P}(\{X = x\} \cap \{Y = y\}) = \mathbb{P}(X = x) \cdot \mathbb{P}(Y = y)$$

for all $x \in \mathcal{X}, y \in \mathcal{Y}$. A collection of random variables $X_1, \ldots, X_n \in \mathcal{X}$ are independent if for any subcollection $I \subseteq [n]$, and any $x_i \in \mathcal{X}$, we have

$$\mathbb{P}\Big(\cap_{i \in I} \{X_i = x_i\}\Big) = \prod_{i \in I} \mathbb{P}(X_i = x_i).$$

Notation: We use $X \perp\!\!\!\perp Y$ to say that X and Y are independent. For a collection of random variables $X_1, \ldots, X_n$ with the same distribution and are all mutually independent, we say that $X_1, \ldots, X_n$ are i.i.d. (independent and identically distributed).

Definition 1.7. The **expectation** of a discrete random variable X, denoted $\mathbb{E}[X]$, is defined as

$$\mathbb{E}[X] = \sum_{x \in \mathcal{X}} x P_X(x).$$

Notation: When there is no ambiguity, we use $\mathbb{E}X$ to denote $\mathbb{E}[X]$.

Exercise 1.6: Linearity of expectation

Show that for discrete random variables X and Y and for any constant c,

$$\mathbb{E}[X+Y] = \mathbb{E}[X] + \mathbb{E}[Y].$$
$$\mathbb{E}[cX] = c \cdot \mathbb{E}X.$$

Hint: For the first identity, let event $A_i = \{X = i\}$ and event $B_j = \{Y = j\}$. Start with

$$\mathbb{E}[X+Y] = \sum_i \sum_j (i+j) \cdot \mathbb{P}\left[A_i \cap B_j\right].$$

A useful random variable often used is the indicator (of a set) random variable. The indicator of set $A \subseteq \Omega$ is denoted $\mathbf{1}_A$ and defined as

$$\mathbf{1}_A(\omega) = \begin{cases} 1 & \text{if } \omega \in A, \\ 0 & \text{otherwise.} \end{cases}$$

It is easy to see that

$$\mathbb{E}\mathbf{1}_A = P(A).$$

Theorem 1.4. *Let X be a discrete random variable that takes on only nonnegative integer values. Then,*

$$\mathbb{E}X = \sum_{i=1}^{\infty} \mathbb{P}[X \geq i].$$

Proof. We have

$$\begin{aligned}
\mathbb{E}X = & \sum_{i=1}^{\infty} i \cdot \mathbb{P}[X = i] \\
= & \ \mathbb{P}[X = 1] \\
& + \mathbb{P}[X = 2] + \mathbb{P}[X = 2] \\
& + \mathbb{P}[X = 3] + \mathbb{P}[X = 3] + \mathbb{P}[X = 3] \\
& + \cdots \\
= & \ \mathbb{P}[X \geq 1] + \mathbb{P}[X \geq 2] + \cdots \\
= & \sum_{i=1}^{\infty} \mathbb{P}[X \geq i].
\end{aligned}$$

where in the penultimate step we summed column-wise; that is,

$$\mathbb{P}[X \geq i] = \mathbb{P}[X = i] + \mathbb{P}[X = i+1] + \cdots.$$

□

Exercise 1.7: Geometric random variable

Show that for a geometric random variable X with $P_X(i) = (1-p)^{i-1}p$ for $i \geq 1$,

$$\mathbb{E}X = \frac{1}{p}.$$

Hint: Use Theorem 1.4.

Example 1.1. *(Coupon Collector's problem)* Suppose that each box of chocolates contains one of n different coupons. Once you obtain one of every type of coupon, you can send in for a prize. Assuming that the coupon in each box is chosen independently and uniformly at random from the n possibilities and that you do not collaborate with others to collect coupons, how many boxes of chocolates must you buy before you obtain at least one of every type of coupon?

Let T_n be the smallest number of boxes that need to be bought to obtain at least one each of the n different coupons. If X_i is the number of boxes purchased while we had exactly $i-1$ different coupons, then

$$X = \sum_{i=1}^{n} X_i.$$

Observe that X_i is a Geometric random variable with parameter $\frac{n-i+1}{n}$ and hence

$$\begin{aligned}
\mathbb{E}X &= \sum_{i=1}^{n} \mathbb{E}X_i \\
&= \sum_{i=1}^{n} \frac{n}{n-i+1} \\
&= n \sum_{i=1}^{n} \frac{1}{i} \\
&\approx n \ln n.
\end{aligned}$$

where the last step is left as an exercise.

Exercise 1.8: Harmonic number

Prove: The harmonic number $H(n) = \sum_{i=1}^{n} \frac{1}{i}$ satisfies $H(n) = \ln n + \Theta(1)$.
Hint: Observe that

$$\ln n = \int_{x=1}^{n} \frac{1}{x} dx \leq \sum_{k=1}^{n} \frac{1}{k}$$

and

$$\sum_{k=2}^{n} \frac{1}{k} \leq \int_{x=1}^{n} \frac{1}{x} dx = \ln n.$$

Draw the graph of $1/x$, $1/\lfloor x \rfloor$, and $1/\lceil x \rceil$ to aid the intuition.

Definition 1.8. The **k-th moment** of a random variable is defined as

$$\mathbb{E}X^k := \sum_{x \in \mathcal{X}} x^k P_X(x).$$

Definition 1.9. The **covariance** of random variables X and Y is given by

$$\mathrm{Cov}(X,Y) := \mathbb{E}\left[(X-\mathbb{E}X)(Y-\mathbb{E}Y)\right].$$

X and Y are said to be **uncorrelated** if $\mathrm{Cov}(X,Y)=0$.

Definition 1.10. The **variance** of a random variable X is defined as

$$\mathrm{Var}(X) := \mathrm{Cov}(X,X).$$

and **standard deviation** of X is given by $\sqrt{\mathrm{Var}(X)}$.

Exercise 1.9: Variance and covariance

1. Show that
$$\mathrm{Cov}(X,Y) = \mathbb{E}[XY] - \mathbb{E}X\cdot\mathbb{E}Y$$
and
$$\mathrm{Var}(X) = \mathbb{E}X^2 - (\mathbb{E}X)^2.$$
2. Let $X_1,\ldots,X_n$ be random variables. Verify:
$$\mathrm{Var}\left(\sum_{i=1}^n X_i\right) = \sum_{i=1}^n \mathrm{Var}(X_i) + 2\sum_{i<j}\mathrm{Cov}(X_i,X_j).$$
What if $X_1,\ldots,X_n$ are mutually independent?
3. Check if the following is true: for a constant c, $\mathrm{Var}(cX) = c\mathrm{Var}(X)$.

Hint: Use the definition of covariance and linearity of expectation.

Definition 1.11. The **conditional expectation** of a random variable Y given $Z=z$ is given by

$$\mathbb{E}[Y|Z=z] = \sum_{y\in\mathcal{Y}} y\cdot\mathbb{P}[Y=y|Z=z].$$

Notice that $P(\cdot|Z=z)$ defines a valid probability distribution on $\mathcal{Y}$ (see Theorem 1.1) and the conditional expectation above is calculated with respect to this conditional distribution.

Example 1.2. Suppose we independently roll a standard six-sided die twice. Let X_1 be the outcome of the first roll and X_2 that of the second. Let X be the sum of the outcomes. Then,

$$\mathbb{E}[X|X_1=2] = \sum_{x=3}^{8} x\cdot\mathbb{P}[X=x|X_1=2] = \frac{3+4+\ldots+8}{6} = \frac{11}{2}$$

and

$$\begin{aligned}\mathbb{E}[X_1|X=5] &= \sum_{x=1}^{4} x\cdot\mathbb{P}[X_1=x|X=5]\\ &= \sum_{x=1}^{4} x\cdot\frac{\mathbb{P}\left[\{X_1=x\}\cap\{X=5\}\right]}{\mathbb{P}[X=5]}\\ &= \sum_{x=1}^{4} x\cdot\frac{1/36}{4/36} = \frac{5}{2}.\end{aligned}$$

Lemma 1.1. *For random variables X and Y,*

$$\mathbb{E}X = \sum_y \mathbb{P}[Y = y]\mathbb{E}[X|Y = y]$$

Proof. We have

$$\begin{aligned}\sum_y \mathbb{P}[Y = y]\mathbb{E}[X|Y = y] &= \sum_y \mathbb{P}[Y = y] \sum_x x \cdot \mathbb{P}[X = x|Y = y] \\ &= \sum_y \sum_x x \cdot \mathbb{P}[X = x, Y = y] \\ &= \sum_x x \cdot \mathbb{P}[X = x] \\ &= \mathbb{E}X.\end{aligned}$$

□

Notice that $\mathbb{E}[Y|Z = z]$ is a function of z. Hence, $\mathbb{E}[Y|Z]$ is a random variable $f(Z)$ that takes the value $\mathbb{E}[Y|Z = z]$ with probability $\mathbb{P}[Z = z]$. Similarly, the **conditional variance**, given by

$$\mathrm{Var}(Y|Z) = \mathbb{E}\left[(Y - \mathbb{E}[Y|Z])^2|Z\right]$$

is also a random variable.

Exercise 1.10: Eden garden chronicles

Prove: For random variables Y and Z,

1. Adam's law:
$$\mathbb{E}Y = \mathbb{E}\Big[\mathbb{E}\left[Y|Z\right]\Big].$$
Hint: Use Lemma 1.1.

2. Eve's law:
$$\mathrm{Var}(Y) = \mathbb{E}\Big[\mathrm{Var}(Y|Z)\Big] + \mathrm{Var}\Big(\mathbb{E}[Y|Z]\Big).$$

Can you guess why the laws are so named? *Hint:* It started with Eve's law.

1.3 Some useful inequalities

1. ***Cauchy-Schwarz inequality:*** For random variables X and Y, we have

$$\mathbb{E}[XY]^2 \leq \mathbb{E}X^2 \cdot \mathbb{E}Y^2.$$

Proof. For $t \in \mathbb{R}$, let

$$\begin{aligned}f(t) &= \mathbb{E}[(tX + Y)^2] \\ &= t^2\mathbb{E}X^2 + 2t\mathbb{E}[XY] + \mathbb{E}Y^2.\end{aligned}$$

Since $f(t) \geq 0$, the discriminant cannot be positive. Therefore,

$$\Big(2\mathbb{E}[XY]\Big)^2 - 4 \cdot \mathbb{E}X^2\mathbb{E}Y^2 \leq 0$$

which completes the proof. □

2. ***Markov's inequality:*** For $X \geq 0$ and $t > 0$, we have

$$\mathbb{P}[X \geq t] \leq \frac{\mathbb{E}X}{t}.$$

Proof. We have

$$t \cdot \mathbf{1}_{\{X \geq t\}} \leq X.$$

Taking expectation on both sides yields

$$t \cdot \mathbb{P}[X \geq t] \leq \mathbb{E}X$$

which completes the proof. □

3. ***Chebyshev's inequality:*** For X with finite mean and variance, and $t > 0$, we have

$$\mathbb{P}\left[|X - \mathbb{E}X| \geq t\right] \leq \frac{\mathrm{Var}(X)}{t^2}.$$

Proof. Apply Markov's inequality to $|X - \mathbb{E}X|^2$. □

Exercise 1.11: How many standard deviations away from the mean?

Show that for X with mean μ and variance σ^2, and $k > 0$, we have

$$\mathbb{P}\left[|X - \mu| \geq k\sigma\right] \leq \frac{1}{k^2}.$$

Example 1.3. *(An estimation problem)* We have deployed n sensors to estimate the Air Quality Index (AQI) of a certain area. Let μ be the actual AQI value and the sensors give out noisy measurements of μ. We model this as: the measurement of sensor i is given by

$$X_i = \mu + Z_i$$

where Z_i are i.i.d. with zero mean and finite variance, σ^2. The goal is to estimate μ using X^n. We use the "natural estimate", namely,

$$\widehat{\mu} = \frac{1}{n}\sum_{i=1}^{n} X_i.$$

How many sensors should be deployed to estimate the AQI value to within an accuracy of $\sigma/10$ with high probability, say 0.95?

Solution: By Chebyshev's inequality, we have

$$\mathbb{P}\left[|\widehat{\mu} - \mu| \geq \frac{\sigma}{10}\right] \leq \frac{100}{n}. \qquad \text{(Verify this!)}$$

Thus, to ensure that

$$\mathbb{P}\left[|\widehat{\mu} - \mu| < \frac{\sigma}{10}\right] \geq 0.95,$$

it suffices to have

$$\frac{100}{n} \leq 0.05.$$

That is, it suffices to deploy $n \geq 2000$ sensors.

Remark – Observe that if we can improve the bound $100/n$ given by Chebyshev's inequality, we can give the same guarantee (on the accuracy and the probability) using fewer sensors. We will develop techniques to achieve this in the next chapter.

4. ***Paley-Zygmund inequality:*** For $X \geq 0$ and $0 < \alpha < 1$, we have

$$\mathbb{P}\Big[X \geq \alpha \mathbb{E}X\Big] \geq (1-\alpha)^2 \frac{(\mathbb{E}X)^2}{\mathbb{E}X^2}.$$

Also,

$$\mathbb{P}[X > 0] \geq \frac{(\mathbb{E}X)^2}{\mathbb{E}X^2}.$$

Proof. We prove the special case first. We have

$$X = X \cdot \mathbf{1}_{\{X>0\}}.$$

Taking expectation on both sides and using Cauchy-Schwarz inequality, we get

$$(\mathbb{E}X)^2 \leq \mathbb{E}X^2 \cdot \mathbb{P}[X > 0].$$

For the general case (first part), observe that

$$X = X \cdot \mathbf{1}_{\{X \geq \alpha \mathbb{E}X\}} + X \cdot \mathbf{1}_{\{X < \alpha \mathbb{E}X\}}$$

Taking expectation on both sides yields

$$\begin{aligned} \mathbb{E}X &= \mathbb{E}\left[X \cdot \mathbf{1}_{\{X \geq \alpha \mathbb{E}X\}}\right] + \mathbb{E}\left[X \cdot \mathbf{1}_{\{X < \alpha \mathbb{E}X\}}\right] \\ &\leq \mathbb{E}\left[X \cdot \mathbf{1}_{\{X \geq \alpha \mathbb{E}X\}}\right] + \alpha \mathbb{E}X \end{aligned}$$

whereby we have

$$(1-\alpha)\mathbb{E}X \leq \mathbb{E}\left[X \cdot \mathbf{1}_{\{X \geq \alpha \mathbb{E}X\}}\right].$$

Applying Cauchy-Schwarz inequality to the right-side completes the proof. □

Exercise 1.12: Application of the inequalities

A fair coin is tossed n times.

(a) Show that the probability of obtaining more than $\frac{1}{\sqrt{2}}$ fraction of heads is at most $\frac{1}{\sqrt{2}}$.

(b) Show that the probability of obtaining more than $\frac{1}{4}$ fraction of heads is at least $\frac{1}{4}$ as $n \to \infty$.

1.3.1 First and Second moment methods

Recall the coupon collector's problem (Example 1.1). We use Markov's inequality in conjunction with Paley-Zygmund inequality to show that the number of boxes T_n "concentrates" around $n \ln n$. We need the following inequalities.

Exercise 1.13: From $1 - x$ to $\exp\{-x\}$

Prove the following:

$$\begin{aligned} 1 - x &\leq e^{-x} && ; x \in \mathbb{R} \\ 1 - x &\geq e^{-x-x^2} && ; |x| \leq 1/2. \end{aligned}$$

Theorem 1.5. *Fix any $\delta > 0$. Then,*

$$\lim_{n \to \infty} \mathbb{P}\left[(1-\delta) n \ln n \leq T_n \leq (1+\delta) n \ln n\right] = 1.$$

Proof. Let

$$X_{t,k} = \mathbf{1}_{\{k\text{-th coupon not seen in the first } t \text{ draws}\}}$$

and let

$$X_t = \sum_{k=1}^{n} X_{t,k}.$$

That is, X_t is the total number of coupons unseen till t draws. Therefore,

$$T_n > t \iff X_t \geq 1.$$

Next, we show that $\mathbb{P}[X_t \geq 1] \to 0$ for $t = (1+\delta)n \ln n$. By Markov's inequality, we have

$$\begin{aligned}
\mathbb{P}[X_t \geq 1] &\leq \mathbb{E}X_t \\
&= \sum_{k=1}^{n} \mathbb{E}X_{t,k} \\
&= n\left(1 - \frac{1}{n}\right)^t \\
&\leq ne^{-\frac{t}{n}}
\end{aligned}$$

where the last step follows by the first inequality in Exercise 1.13. Letting $t = (1+\delta)n \ln n$, we have

$$\mathbb{P}[X_t \geq 1] \leq n^{-\delta}$$

which goes to 0 as $n \to \infty$. Next, we show that $\mathbb{P}[X_t \geq 1] \to 1$ for $t = (1-\delta)n \ln n$. By Paley-Zygmund inequality, we have

$$\begin{aligned}
\mathbb{P}[X_t \geq 1] &= \mathbb{P}[X_t > 0] \\
&\geq \frac{(\mathbb{E}X_t)^2}{\mathbb{E}X_t^2}.
\end{aligned}$$

We have

$$\begin{aligned}
X_t^2 &= (X_{t,1} + \cdots + X_{t,n})^2 \\
&= \sum_{k=1}^{n} X_{t,k}^2 + 2 \sum_{1 \leq k \leq \ell \leq n} X_{t,k} X_{t,\ell}
\end{aligned}$$

whereby

$$\mathbb{E}X_t^2 = \sum_{k=1}^{n} \mathbb{E}X_{t,k} + 2 \sum_{1 \leq k \leq \ell \leq n} \mathbb{E}\left[X_{t,k} X_{t,\ell}\right]$$

where we have used $X_{t,k}^2 = X_{t,k}$. Observe that $\mathbb{E}\left[X_{t,k} X_{t,\ell}\right]$ is the probability that both k-th and ℓ-th coupon are unseen in the first t draws. Therefore,

$$\mathbb{E}X_t^2 = n\left(1 - \frac{1}{n}\right)^t + n(n-1)\left(1 - \frac{2}{n}\right)^t.$$

and hence

$$\begin{aligned}
\mathbb{P}[X_t \geq 1] &\geq \frac{\left(n\left(1 - \frac{1}{n}\right)^t\right)^2}{n\left(1 - \frac{1}{n}\right)^t + n(n-1)\left(1 - \frac{2}{n}\right)^t} \\
&\geq \frac{n^2 e^{-\frac{2t}{n} - \frac{2t}{n^2}}}{ne^{-\frac{t}{n}} + n(n-1)e^{-\frac{2t}{n}}}
\end{aligned}$$

where the last step follows by the inequalities in Exercise 1.13 (plug in and check!). Letting $t = (1-\delta)n \ln n$ and letting $n \to \infty$ allows us to conclude that $\mathbb{P}[X_t \geq 1] \to 1$. $\square$

1.4 Limit Theorems

Limit theorems state roughly that as we collect more and more samples, we start seeing "some pattern" in the average of the samples. For instance, the law of large numbers formalizes what we know as folklore - *sample average converges to the statistical average*, and the Central Limit Theorem[2] tells us that *the sum of a large number of random variables, with appropriate scaling, behaves roughly like a Gaussian random variable.*

Let $X_1, \ldots, X_n$ be i.i.d. with common distribution P and let

$$S_n = X_1 + \cdots + X_n.$$

Let $\mathbb{E}X_1 = \mu < \infty$ and $\mathrm{Var}(X_1) = \sigma^2 < \infty$.

Exercise 1.14: Sum and average

Verify: $\mathbb{E}S_n = n\mu$, $\mathrm{Var}(S_n) = n\sigma^2$, $\mathbb{E}\left[\frac{S_n}{n}\right] = \mu$, $\mathrm{Var}\left(\frac{S_n}{n}\right) = \frac{\sigma^2}{n}$.

1. ***Weak Law of Large Numbers (WLLN):*** For any $\delta > 0$, we have

$$\lim_{n\to\infty} \mathbb{P}\left(\left|\frac{S_n}{n} - \mu\right| > \delta\right) = 0.$$

Proof. By Chebyshev's inequality, We have

$$\mathbb{P}\left(\left|\frac{S_n}{n} - \mu\right| > \delta\right) \leq \frac{\mathrm{Var}\left(\frac{S_n}{n}\right)}{\delta^2} = \frac{\sigma^2}{n\delta^2} \to 0 \text{ as } n \to \infty.$$

□

2. ***Strong Law of Large Numbers (SLLN):*** S_n/n converges to μ *almost surely.* That is,

$$P\left(\omega : \lim_{n\to\infty} \frac{S_n}{n} = \mu\right) = 1.$$

3. ***Lindeberg-Lévy Central Limit Theorem (CLT):*** Let Z_n denote the centered, normalized S_n (i.e., make the mean and variance of the resulting random variable 0 and 1, respectively). That is,

$$Z_n = \frac{S_n - \mathbb{E}S_n}{\sqrt{\mathrm{Var}(S_n)}} = \frac{S_n - n\mu}{\sigma\sqrt{n}}.$$

Then, as $n \to \infty$,

$$Z_n \to \mathcal{N}(0,1) \quad \textit{in distribution.}$$

That is, for any $t \in \mathbb{R}$ and $G \sim \mathcal{N}(0,1)$,

$$\mathbb{P}[Z_n \geq t] \to \mathbb{P}[G \geq t] = \frac{1}{\sqrt{2\pi}} \int_t^\infty e^{-x^2/2} dx.$$

4. ***de Moivre-Laplace theorem:*** If X_i are i.i.d. $\mathrm{Ber}(p)$, then as $n \to \infty$,

$$\frac{S_n - np}{\sqrt{np(1-p)}} \to \mathcal{N}(0,1) \quad \textit{in distribution};$$

essentially, CLT for the special case of $\mathrm{Ber}(p)$ random variables.

[2] As the name suggests, this result is a *central*/important limit theorem in probability and statistics.

Exercise 1.1: Reading exercise

We have used three different notions of convergence in WLLN, SLLN, and CLT (convergence in probability, almost sure convergence, and convergence in distribution, respectively). Find out what these are. Are there other notions of convergence of random variables?

Armed with these limit theorems, let us try and answer the following question:

Example 1.4. We toss a fair coin n times. What is the probability that we get at least $\frac{3n}{4}$ heads?

For $i \in [n]$, for the i-th toss, let $X_i = \mathbf{1}_{\{H\}}$. Then, $S_n = \sum_{i=1}^n X_i$ is the total number of heads in n tosses, $\mathbb{E}S_n = n/2$ and $\mathrm{Var}(S_n) = n/4$. We seek to upper bound the probability that S_n exceeds $\frac{3n}{4}$. The tighter (smaller) this bound, the better.

(a) No-brainer inequality: We have

$$\mathbb{P}\left[S_n \geq \frac{3n}{4}\right] \leq 1.$$

(b) Markov's inequality: We have

$$\mathbb{P}\left[S_n \geq \frac{3n}{4}\right] \leq \frac{n/2}{3n/4} = \frac{2}{3}.$$

(c) Chebyshev's inequality: We have

$$\begin{aligned}\mathbb{P}\left[S_n \geq \frac{3n}{4}\right] &\leq \mathbb{P}\left[\left|S_n - \frac{n}{2}\right| \geq \frac{n}{4}\right] \\ &\leq \frac{\mathrm{Var}\left(S_n\right)}{n^2/16} \\ &= \frac{4}{n}.\end{aligned}$$

Observe that here the bound is shrinking as n increases and hence is better than the bound given by Markov's inequality.

(d) CLT: We have

$$\begin{aligned}\mathbb{P}\left[S_n \geq \frac{3n}{4}\right] &= \mathbb{P}\left[\frac{S_n - n/2}{\sqrt{n/4}} \geq \sqrt{\frac{n}{4}}\right] \\ &= \mathbb{P}\left[Z_n \geq \sqrt{\frac{n}{4}}\right] \\ &\approx \mathbb{P}\left[G \geq \sqrt{\frac{n}{4}}\right].\end{aligned}$$

We use the following bound on the complementary CDF of G.

Exercise 1.15: Gaussian tails

Verify that for all $t > 0$,

$$\left(\frac{1}{t} - \frac{1}{t^3}\right)\frac{1}{\sqrt{2\pi}}e^{-t^2/2} \leq \mathbb{P}[G \geq t] \leq \frac{1}{t}\frac{1}{\sqrt{2\pi}}e^{-t^2/2}.$$

In particular, for $t \geq 1$, we have

$$\mathbb{P}[G \geq t] \leq \frac{1}{\sqrt{2\pi}}e^{-t^2/2}.$$

Then, for $n \geq 4$,

$$\mathbb{P}\left[S_n \geq \frac{3n}{4}\right] \leq \frac{1}{\sqrt{2\pi}} e^{-n/8}.$$

This is **much better** than the bound given by Chebyshev's inequality; however, it is not exact since we used an approximation, guided by CLT. It turns out that the error in this approximation is of the order of $1/\sqrt{n}$ and hence the "exponential decay" we obtained is sidelined/dominated by the error term. The following result formally gives the approximation error.

5. ***Berry-Esseen theorem:*** We have

$$\left|\mathbb{P}[Z_n \geq t] - \mathbb{P}[G \geq t]\right| \leq \frac{c}{\sqrt{n}}$$

for some constant $c > 0$ that depends only on the distribution of X_i. One might wonder: This is only an upper bound on the error; can it be improved? We provide an example that shows that it cannot be, in general.

Example 1.5. Let S_n denote the number of heads in n independent fair coin tosses. Then,

$$\mathbb{P}\left[S_n = \frac{n}{2}\right] = 2^{-n}\binom{n}{n/2}$$

which by Striling's approximation (check it out!) is roughly $c'/\sqrt{n}$. However, observe that

$$\mathbb{P}\left[S_n = \frac{n}{2}\right] = \mathbb{P}[Z_n = 0]$$

whereas

$$\mathbb{P}[G = 0] = 0.$$

Therefore, the error between the two probabilities is of the order of $1/\sqrt{n}$.

This motivates the question: Can we obtain precise bounds similar to that given by the CLT approximation? The study of concentration inequalities answers this in the affirmative.

Chapter 2

Concentration inequalities

Surprisingly, the innocent-looking Markov's inequality itself can be used to obtain precise exponential bounds *à la* that given by the CLT approximation. We derive this class of bounds, commonly called *Chernoff bounds* next. Observe that for any random variable X, any $t \in \mathbb{R}$, and $\lambda > 0$, we have

$$\begin{aligned}\mathbb{P}[X \geq t] &= \mathbb{P}[\lambda X \geq \lambda t] \\ &= \mathbb{P}\left[e^{\lambda X} \geq e^{\lambda t}\right] \\ &\leq \mathbb{E}\left[e^{\lambda X}\right] e^{-\lambda t}\end{aligned}$$

where the last step is by Markov's inequality since $e^{\lambda X}$ is a nonnegative random variable, provided that $\mathbb{E}\left[e^{\lambda X}\right]$, the moment generating function of X exists. Since we are free to choose $\lambda > 0$, we obtain the tightest bound (using this method) as:

$$\mathbb{P}[X \geq t] \leq \min_{\lambda > 0} \mathbb{E}\left[e^{\lambda X}\right] e^{-\lambda t}.$$

Notation: We use $M_X(\lambda)$ to denote the MGF of X evaluated at λ.

Exercise 2.1: Properties of MGF

1. Let $X_i, i \in [n]$ be mutually independent and let $X = \sum_{i=1}^n X_i$. Show that

$$M_X(\lambda) = \prod_{i=1}^n M_{X_i}(\lambda).$$

2. Compute the MGF of $X \sim \mathrm{Ber}(p)$ and $Y \sim \mathrm{Bin}(n, p)$.
   Hint: For the latter, use the first part of the exercise.

2.1 Hoeffding's inequality

Hoeffding's inequality provides exponential tail bounds for bounded random variables. We develop the essential ideas by first deriving Chernoff bounds for binary random variables.

2.1.1 Rademacher random variables

Let X be a binary random variable with

$$\mathbb{P}[X = 1] = 1 - \mathbb{P}[X = 1] = \frac{1}{2};$$

X is called a *Rademacher* random variable.

Theorem 2.1. *Let $X_1, \ldots, X_n$ be i.i.d. Rademacher random variables and let $a = (a_1, \ldots, a_n) \in \mathbb{R}^n$. Then, for any $t \geq 0$, we have*

$$\mathbb{P}\left[\sum_{i=1}^n a_i X_i \geq t\right] \leq \exp\left\{-\frac{t^2}{2\|a\|_2^2}\right\}.$$

Proof. Assume without loss of generality that $\|a\|_2 = 1$ (why?). The Chernoff bound is given by

$$\begin{aligned}\mathbb{P}\left[\sum_{i=1}^n a_i X_i \geq t\right] &= \mathbb{P}\left[\exp\left\{\lambda \sum_{i=1}^n a_i X_i\right\} \geq \exp\{\lambda t\}\right] \\ &\leq e^{-\lambda t} \cdot \mathbb{E}\left[e^{\lambda \sum_{i=1}^n a_i X_i}\right].\end{aligned}$$

The MGF of $\sum_{i=1}^n a_i X_i$ is the product of MGFs of $a_i X_i$ which is given by

$$\begin{aligned}M_{a_i X_i}(\lambda) &= \mathbb{E}\left[e^{a_i X_i \lambda}\right] \\ &= \frac{1}{2}e^{a_i \lambda} + \frac{1}{2}e^{-a_i \lambda} \\ &\leq e^{a_i^2 \lambda^2/2}\end{aligned}$$

where the last step is left as an exercise.

Exercise 2.2: Bounding the cosh function

Show that for all $x \in \mathbb{R}$, we have

$$\frac{1}{2}e^x + \frac{1}{2}e^{-x} \leq e^{x^2/2}.$$

Hint: Use the Taylor series expansion of the exponential function and the inequality (verify this!)

$$(2n)! \geq 2^n \cdot n!.$$

Then, we have

$$\begin{aligned}\mathbb{P}\left[\sum_{i=1}^n a_i X_i \geq t\right] &\leq e^{-\lambda t} \cdot \prod_{i=1}^n e^{a_i^2 \lambda^2/2} \\ &= \exp\left\{-\lambda t + \frac{\lambda^2}{2}\sum_{i=1}^n a_i^2\right\} \\ &= \exp\left\{-\lambda t + \frac{\lambda^2}{2}\right\} \qquad \text{(since } \|a\|_2^2 = a_1^2 + \cdots + a_n^2 = 1\text{)}.\end{aligned}$$

Minimizing the right-side over $\lambda > 0$ yields the desired bound. □

Exercise 2.3: Two-sided tails

Show that

$$\mathbb{P}\left[\left|\sum_{i=1}^n a_i X_i\right| \geq t\right] \leq 2\exp\left\{-\frac{2t^2}{\|a\|_2^2}\right\}.$$

Hint: Use

$$\mathbb{P}\left[\left|\sum_{i=1}^n a_i X_i\right| \geq t\right] = \mathbb{P}\left[\sum_{i=1}^n a_i X_i \geq t\right] + \mathbb{P}\left[-\sum_{i=1}^n a_i X_i \geq t\right]$$

and observe that $-X_i$ has the same distribution as X_i.

2.1.2 Bernoulli random variables

Next, we adapt Theorem 2.1 to the sum of i.i.d. Bernoulli random variables.

Theorem 2.2. *Let $Y_1, \ldots, Y_n$ be i.i.d.* $\mathrm{Ber}(1/2)$ *random variables. Let $\mu = n/2$ denote the mean of the sum $Y = \sum_{i=1}^n Y_i$. Then, for any $t \geq 0$, we have*

$$\mathbb{P}\left[Y \geq \mu + t\right] \leq \exp\left\{-\frac{2t^2}{n}\right\}$$

and for any $0 < \delta < 1$, we have

$$\mathbb{P}\left[Y \geq (1+\delta)\mu\right] \leq \exp\left\{-\frac{n\delta^2}{2}\right\}.$$

Proof. For $X_i, i \in [n]$ i.i.d. Rademacher random variables, let $X = \sum_{i=1}^n X_i$. Since we have $Y_i = \frac{X_i+1}{2}$ and as a result,

$$Y = \sum_{i=1}^n Y_i = \frac{1}{2}\sum_{i=1}^n X_i + \frac{n}{2} = \frac{X}{2} + \frac{n}{2},$$

we can write

$$\begin{aligned}\mathbb{P}[Y \geq \mu + t] &= \mathbb{P}\left[X \geq 2t\right] && \text{(since } \mu = n/2\text{)}\\ &\leq \exp\left\{-\frac{2t^2}{n}\right\}.\end{aligned}$$

by Theorem 2.1. The proof of the other inequality is similar and is left as an exercise. □

Exercise 2.4: Two-sided tails and generalization

1. Prove the second inequality in Theorem 2.2 and extend the result to two-sided tails as in Exercise 2.3.

2. State and prove Theorem 2.1 with $X_i - \mathbb{E}X_i$ in place of X_i when

$$\mathbb{P}[X_i = 1] = p_i = 1 - \mathbb{P}[X_i = -1]$$

and derive the result corresponding to Theorem 2.2 for $Y_i \sim \mathrm{Ber}(p)$.
Hint: For the former, you may assume the inequality: for all $\lambda > 0$ and $0 < p < 1$,

$$pe^{2(1-p)\lambda} + (1-p)e^{-2p\lambda} \leq e^{\lambda^2/2}.$$

Example 2.1. *(Revisiting the coin toss example)* Recall Example 1.4: We toss a fair coin n times. What is the probability that we get at least $\frac{3n}{4}$ heads?

For $i \in [n]$, for the i-th toss, let $X_i = \mathbf{1}_{\{H\}}$. Then, $S_n = \sum_{i=1}^n X_i$ is the total number of heads in n tosses and $\mathbb{E}S_n = n/2$. Using Theorem 2.2 we have

$$\begin{aligned}\mathbb{P}\left[S_n \geq \frac{3n}{4}\right] &= \mathbb{P}\left[S_n \geq \frac{n}{2} + \frac{n}{4}\right]\\ &\leq \exp\left\{-\frac{2(n/4)^2}{n}\right\}\\ &= e^{-n/8}\end{aligned}$$

which (order-wise) matches the bound yielded by the CLT approximation.

Example 2.2. *(Boosting randomized algorithms)* We have an algorithm for solving a decision problem. The algorithm makes a decision at random and returns the correct answer with probability $\frac{1}{2}+\delta$ for some fixed $\delta > 0$. (Observe that this is better than an algorithm that randomly guesses yes/no). To improve the performance, suppose we run the algorithm n times and take the majority vote. How large should n be to guarantee that we output the correct answer with probability at least $1-\varepsilon$ for a fixed $\varepsilon > 0$?

Solution: Let

$$X_i = \mathbf{1}_{\{\text{the answer in the } i\text{-th run is wrong}\}}$$

and let $X = \sum_{i=1}^n X_i$. Then, we have

$$\mathbb{E}X_i = \frac{1}{2} - \delta \quad \text{and} \quad \mathbb{E}X = n\left(\frac{1}{2} - \delta\right).$$

We output the wrong answer if $X \geq n/2$. This happens with probability

$$\begin{aligned}\mathbb{P}[X \geq n/2] &= \mathbb{P}[X \geq \mathbb{E}X + n\delta] \\ &\leq \exp\left\{-\frac{2n^2\delta^2}{n}\right\} \\ &= \exp\left\{-2n\delta^2\right\}\end{aligned}$$

where the inequality is by Theorem 2.2. Hence, to ensure correctness with probability at least $1-\varepsilon$, it suffices to have

$$n \geq \frac{1}{2\delta^2} \ln \frac{1}{\varepsilon}.$$

Exercise 2.5: Lower bound on correctness

Solve the problem in Example 2.2 when the algorithm *returns the correct answer with probability* ***at least*** $\frac{1}{2} + \delta$ *for some fixed* $\delta > 0$.

Example 2.3. *(Median trick)* Let $\widehat{\mu}(X)$ be an estimator of a parameter μ, based on sample X that satisfies

$$\mathbb{P}\left[|\mu - \widehat{\mu}(X)| > \delta\right] \leq \frac{1}{3}.$$

Then, for $X_1, \ldots, X_n$ that are n independent copies of X, there exists an estimator $\widehat{\mu_n}(X^n)$ that satisfies

$$\begin{aligned}&(i) \quad \mathbb{P}\big[\,|\mu - \widehat{\mu_n}(X^n)| > \delta\big] \leq \varepsilon, \\ &(ii) \quad n = O\left(\log \frac{1}{\varepsilon}\right).\end{aligned}$$

Proof. Let

$$\widehat{\mu_n}(X_1, \ldots, X_n) = \text{median}\left\{\widehat{\mu}(X_1), \ldots, \widehat{\mu}(X_n)\right\}.$$

Then,

$$|\mu - \widehat{\mu_n}(X^n)| > \delta \Longrightarrow |\{i : |\mu - \widehat{\mu}(X_i)| > \delta\}| \geq \frac{n}{2}. \tag{2.1}$$

Let

$$B_i = \mathbf{1}_{\{|\mu - \widehat{\mu}(X_i)| > \delta\}}.$$

Then, $B_i \sim \text{Ber}(p_i)$ where $p_i \leq 1/3$. Thus, by (2.1), we have

$$\begin{aligned}
\mathbb{P}\left[|\mu - \widehat{\mu_n}(X^n)| > \delta\right] &\leq \mathbb{P}\left[\sum_{i=1}^n B_i \geq \frac{n}{2}\right] \\
&= \mathbb{P}\left[\sum_{i=1}^n (B_i - p_i) \geq \frac{n}{2} - \sum_{i=1}^n p_i\right] \\
&\leq \mathbb{P}\left[\sum_{i=1}^n (B_i - p_i) \geq \frac{n}{2} - \frac{n}{3}\right] \\
&\leq \exp\left\{-\frac{n}{18}\right\}
\end{aligned}$$

by Theorem 2.2. Hence, in order to ensure that (i) is satisfied, it suffices to have

$$n \geq 18 \ln \frac{1}{\varepsilon}$$

thereby satisfying (ii) as well. □

2.1.3 Bounded random variables

Observe that the proof of Theorem 2.1 hinges upon upper-bounding the MGF using an appropriate exponential function. We derive such a result for random variables bounded between m and M first and then use it to show concentration bounds for such random variables, akin to Thereom 2.1. This is the result commonly known as Hoeffding's inequality.

Lemma 2.1 (Hoeffding's lemma)**.** *For a random variable X such that $m \leq X \leq M$, for $\lambda > 0$, we have*

$$\mathbb{E}\left[e^{\lambda(X - \mathbb{E}X)}\right] \leq \exp\left\{\frac{\lambda^2 (M-m)^2}{8}\right\}.$$

Proof. Let $Y = X - \mathbb{E}X$ and let

$$\psi_Y(\lambda) := \ln \mathbb{E} e^{\lambda Y}$$

the log MGF of Y. Then, (verify this!)

$$\psi_Y(0) = \psi_Y'(0) = 0 \tag{2.2}$$

and

$$\psi_Y''(\lambda) = e^{-\psi_Y(\lambda)} \mathbb{E}\left[Y^2 e^{\lambda Y}\right] - e^{-2\psi_Y(\lambda)} \left(\mathbb{E}\left[Y e^{\lambda Y}\right]\right)^2.$$

Let Z be a random variable with PDF (see definition 1.6)

$$f_Z(z) = e^{-\psi_Y(\lambda)} e^{\lambda z} f_Y(z)$$

where $f_Y(\cdot)$ is the PDF of Y. Note that

$$\int_z f_Z(z) dz = e^{-\psi_Y(\lambda)} \int_z e^{\lambda z} f_Y(z) = e^{-\psi_Y(\lambda)} \mathbb{E} e^{\lambda Y} = 1$$

and hence f_Z is a valid PDF.

Exercise 2.6: Discrete case

Specialize the construction of Z above to the case where Y is discrete.

Then, for any $\lambda > 0$,

$$\psi_Y''(\lambda) = \mathrm{Var}(Z)$$

and since Z takes values in $[m, M]$, by Popoviciu's inequality (Exercise 2.7), we have

$$\psi_Y''(\lambda) \leq \frac{(M-m)^2}{4}. \tag{2.3}$$

Then, for every $\lambda > 0$, by Taylor's theorem, there exists $\widetilde{\lambda} \in (0, \lambda)$ such that

$$\begin{aligned}\psi_Y(\lambda) &= \psi_Y(0) + \lambda\psi_Y'(0) + \frac{\lambda^2}{2}\psi_Y''\left(\widetilde{\lambda}\right) \\ &\leq \frac{\lambda^2(M-m)^2}{8}\end{aligned}$$

by (2.2) and (2.3). □

Exercise 2.7: Variance bounds

1. For a random variable X with finite expectation, show that $\mathbb{E}X$ is the value of c that minimizes

$$g(c) := \mathbb{E}[(X-c)^2].$$

2. (Popoviciu's inequality) For a random variable X such that $m \leq X \leq M$ where m and M are constants, show that

$$\mathrm{Var}(X) \leq \frac{(M-m)^2}{4}.$$

Hint: Argue using (a) that $g(\mathbb{E}X) \leq g\left(\frac{M+m}{2}\right)$.

3. (Bhatia-Davis inequality) For a random variable $X \in [m, M]$ with mean μ where m and M are constants, show that

$$\mathrm{Var}(X) \leq (M-\mu)(\mu-m).$$

Hint: Use $\mathbb{E}[(M-X)(X-m)] \geq 0$ (or) Cauchy-Schwarz on $\mathbb{E}[(X-\mu)(X-\mu)]$.

4. Prove Popoviciu's inequality using the Bhatia-Davis inequality.

Next, we use Hoeffding's lemma to prove Hoeffding's inequality.

Theorem 2.3 (Hoeffding's inequality)**.** *For $X_1, \ldots, X_n$ independent with $m_i \leq X_i \leq M_i$, for $t > 0$, we have*

$$\mathbb{P}\left[\sum_{i=1}^n (X_i - \mathbb{E}X_i) \geq t\right] \leq \exp\left\{-\frac{2t^2}{\sum_{i=1}^n (M_i - m_i)^2}\right\}.$$

Proof. The Chernoff bound for $\sum_{i=1}^n (X_i - \mathbb{E}X_i)$ is given by

$$\begin{aligned}\mathbb{P}\left[\sum_{i=1}^n (X_i - \mathbb{E}X_i) \geq t\right] &\leq e^{-\lambda t}\mathbb{E}\left[e^{\lambda\sum_{i=1}^n (X_i - \mathbb{E}X_i)}\right] \\ &= e^{-\lambda t}\prod_{i=1}^n \mathbb{E}\left[e^{\lambda(X_i - \mathbb{E}X_i)}\right] \\ &\leq \exp\left\{-\lambda t + \frac{\lambda^2}{8}\sum_{i=1}^n (M_i - m_i)^2\right\}\end{aligned}$$

by Hoeffding's lemma. Setting

$$\lambda = \frac{4t}{\sum_{i=1}^{n}(M_i - m_i)^2}$$

proves the result. □

Exercise 2.8: Hoeffding's inequality

1. Verify that the value of λ chosen in the proof of Theorem 2.3 gives the best possible Chernoff bound.

2. Consider a collection $X_1, \ldots, X_n$ of n integers chosen independently and uniformly from the set $\{0, 1, 2\}$. Let $X = \sum_{i=1}^{n} X_i$ and $0 < \delta < 1$. Derive upper bounds for $\mathbb{P}[X \geq (1+\delta)n]$ and $\mathbb{P}[X \leq (1-\delta)n]$.

Example 2.4. *(Learning in a Finite Hypothesis Space)* We observe samples $(X_1, Y_1), \ldots, (X_m, Y_m)$ i.i.d. D, some fixed distribution and seek to choose a hypothesis $h \in \mathcal{H}$, where $\mathcal{H}$ is a fixed class of hypotheses, such that $h(x) = y$. For $h \in \mathcal{H}$, define the training error upon observing $\{(x_i, y_i\}_{i=1}^{m}$ to be

$$\widehat{e}(h) = \frac{1}{m}\sum_{i=1}^{m} \mathbf{1}_{\{h(x_i) \neq y_i\}}$$

and the generalization error to be

$$e_D(h) = \mathbb{P}[h(X) \neq Y]$$

where $(X, Y) \sim D$. The goal is to ensure that

$$\mathbb{P}\left[\forall h \in \mathcal{H} : |\widehat{e}(h) - e_D(h)| \leq \varepsilon\right] \geq 1 - \delta.$$

How many training samples must we have to ensure this?[1]
Solution: For a fixed $h \in \mathcal{H}$, we have

$$\mathbb{E}\left[\widehat{e}(h)\right] = e_D(h).$$

Therefore, Hoeffding's inequality yields

$$\mathbb{P}\left[|e_D(h) - \widehat{e}(h)| > \varepsilon\right] \leq 2\exp\{-2\epsilon^2 m\}.$$

To ensure that the above inequality holds for all $h \in \mathcal{H}$ with probability $1 - \delta$ it suffices to ensure (by union bound) that

$$\begin{aligned}\mathbb{P}\left[\exists h \in \mathcal{H} : |e_D(h) - \widehat{e}(h)| > \varepsilon\right] &\leq \sum_{h \in \mathcal{H}} \mathbb{P}\left[|e_D(h) - \widehat{e}(h)| > \varepsilon\right] \\ &\leq 2|\mathcal{H}|\exp\{-2\epsilon^2 m\} \\ &\leq \delta.\end{aligned}$$

That is, it suffices to have

$$m \geq \frac{\ln 2|\mathcal{H}| + \ln 1/\delta}{2\varepsilon^2}$$

training samples.

[1] If this property (referred to as *uniform convergence property*) holds, then it can be shown that the generalization error of the empirical risk minimizer (ERM)

$$\widehat{h} = \arg\min_{h \in \mathcal{H}} \widehat{e}(h)$$

is within 2ε of that of the best hypothesis in the hypothesis class, namely,

$$h_{\text{opt}} = \arg\min_{h \in \mathcal{H}} e_D(h).$$

2.2 Sub-Gaussian random variables

Observe that the crucial result used in proving exponential tail bounds for bounded random variables is Hoeffding's lemma. Thus, for any class of random variables for which the MGF satisfies the property given by the lemma, it is possible to derive an exponential tail bound. We call such random variables *sub-Gaussian*.

Definition 2.1. A centered random variable X is said to be **sub-Gaussian** with variance parameter $v > 0$ if the MGF of X satisfies

$$M_X(\lambda) \leq e^{\lambda^2 v/2}$$

for all $\lambda \in \mathbb{R}$. We say that $X \in \mathcal{G}(v)$.

That is, the MGF of a sub-Gaussian random variable with variance parameter σ^2 is upper bounded by the MGF of a Gaussian random variable with zero mean, variance σ^2 (see Exercise 2.9).

Exercise 2.9: Gaussian MGF

Check that for $Z \sim \mathcal{N}(0, \sigma^2)$, we have $M_Z(\lambda) = e^{\lambda^2\sigma^2/2}$.

The following exponential tail bound follows easily.

Theorem 2.4. *For $X \in \mathcal{G}(v)$ and $t > 0$, we have*

$$\mathbb{P}\left[|X| \geq t\right] \leq 2\exp\left\{-\frac{t^2}{2v}\right\}.$$

Proof. The Chernoff bounds are given by

$$\mathbb{P}[X \geq t] \leq \min_{\lambda>0} e^{-\lambda t}\mathbb{E}e^{\lambda X} \leq \min_{\lambda>0} e^{-\lambda t}e^{\lambda^2 v/2} = e^{-t^2/2v}$$

and

$$\mathbb{P}[X \leq -t] \leq \min_{\lambda<0} e^{\lambda t}\mathbb{E}e^{\lambda X} \leq \min_{\lambda<0} e^{\lambda t}e^{\lambda^2 v/2} = e^{-t^2/2v}.$$

The proof is complete by observing that

$$\mathbb{P}\left[|X| \geq t\right] = \mathbb{P}\left[X \geq t\right] + \mathbb{P}\left[X \leq -t\right].$$

□

Exercise 2.10: Properties of sub-Gaussians

Prove:

1. If $X \sim \mathcal{G}(v)$, then
$$\mathrm{Var}(X) \leq v.$$
2. If $X_1, \ldots, X_n$ are independent with $X_i \in \mathcal{G}(v_i)$ and $(a_1, \ldots, a_n) \in \mathbb{R}^n$, then
$$\sum_{i=1}^n a_i X_i \in \mathcal{G}\left(\sum_{i=1}^n a_i^2 v_i\right).$$

Clearly, if $X \sim \mathcal{N}(0, \sigma^2)$ then $X \in \mathcal{G}(\sigma^2)$. It can be shown that bounded random variables with zero mean are also sub-Gaussian.

Exercise 2.11: Examples of sub-Gaussians

1. Show that a Rademacher random variable is sub-Gaussian with variance parameter 1. Use this in conjunction with part 2 of Exercise 2.10 and Theorem 2.4 to solve Exercise 2.3.
2. For X such that $a \leq X \leq b$, exhibit a value of v such that $X \in \mathcal{G}(v)$.

2.3 McDiarmid's inequality

We seek to extend the concentration bounds from sums of independent random variables to ("well-behaved") general functions of independent random variables. As a first step, we consider functions with bounded differences.

Definition 2.2. A function $f : \mathcal{X}^n \to \mathbb{R}$ has the **bounded differences property** (BDP) if for $c_i \geq 0, i \in [n]$,

$$f(x_1, \ldots, x_{i-1}, x_i, x_{i+1}, \ldots, x_n) - f(x_1, \ldots, x_{i-1}, x_i', x_{i+1}, \ldots, x_n) \leq c_i$$

for all $x_1, \ldots, x_n, x_i' \in \mathcal{X}$.

That is, if we change the value of the i-th variable of f while keeping all the others fixed, then the value of the function cannot change by more than c_i.

Theorem 2.5. *Let $X_1, \ldots, X_n$ be independent. Let $f : \mathcal{X}^n \to \mathbb{R}$ satisfy the BDP with $(c_1, \ldots, c_n)$. Then,*

$$\mathbb{P}\left[f(X^n) - \mathbb{E}f(X^n) \geq t\right] \leq \exp\left\{-\frac{t^2}{2\|c\|_2^2}\right\}.$$

Proof. Let

$$\Delta_i := \mathbb{E}\left[f(X^n)|X^i\right] - \mathbb{E}\left[f(X^n)|X^{i-1}\right], \; i \in [n].$$

Then, we have

$$\begin{aligned}
&(i) \quad \sum_{i=1}^n \Delta_i = f(X^n) - \mathbb{E}f(X^n). \\
&(ii) \quad |\Delta_i| \leq c_i. \\
&(iii) \quad \mathbb{E}\Delta_i = \mathbb{E}\Big[\mathbb{E}\left[f(X^n)|X^i\right] - \mathbb{E}\left[f(X^n)|X^{i-1}\right]\Big] \\
&\qquad\qquad = \mathbb{E}f(X^n) - \mathbb{E}f(X^n) = 0
\end{aligned}$$

where in the penultimate step, we used Adam's law (see Exercise 1.10). Further, we make the following claim.

Claim: Δ_is are uncorrelated.

Observe that for deriving Hoeffding's inequality, it suffices for the random variables to be bounded and uncorrelated (rather than independent, although we stated the theorem with independence[2]). Therefore, we have (check this!)

$$\mathbb{P}\left[\sum_{i=1}^n \Delta_i \geq t\right] \leq \exp\left\{-\frac{t^2}{2\sum_{i=1}^n c_i^2}\right\}$$

which completes the proof. □

[2]Recall that the only step in the proof of Theorem 2.3 in which we used independence of $X_1, \ldots, X_n$ was in writing the MGF of the sum $\sum_{i=1}^n (X_i - \mathbb{E}X_i)$ as the product of MGFs; this step is valid if $X_1, \ldots, X_n$ are uncorrelated.

Exercise 2.12: McDiarmid from Hoeffding

Verify the last step in the proof of Theorem 2.5.

Proof of claim. Let

$$Z_i = \mathbb{E}\left[f(X^n)|X^i\right].$$

Then, for $i < j$, we have

$$\begin{aligned}\mathbb{E}\left[Z_j - Z_{j-1}\middle|X^i\right] &= \mathbb{E}\left[Z_j\middle|X^i\right] - \mathbb{E}\left[Z_{j-1}\middle|X^i\right] \\ &= \mathbb{E}\left[\mathbb{E}\left[f(X^n)|X^j\right]\middle|X^i\right] - \mathbb{E}\left[\mathbb{E}\left[f(X^n)|X^{j-1}\right]\middle|X^i\right] \\ &= \mathbb{E}\left[f(X^n)|X^i\right] - \mathbb{E}\left[f(X^n)|X^i\right] \\ &= 0. \end{aligned} \tag{2.4}$$

by the *law of iterated expectations*, a generalization of Adam's law which states that for random variables X, Y, Z,

$$\mathbb{E}\big[\mathbb{E}[Y|X,Z]\big|X\big] = \mathbb{E}[Y|X].$$

Exercise 2.1: Reading exercise

Find out how the law of iterated expectations is stated and proved. A useful analogy for understanding the law is the following (Courtesy – KevinKim on stats.stackexchange):

conditional expectation $\mathbb{E}[Y|X]$ is a picture clicked on a camera with resolution X.

If you click a picture with resolution (X, Z) and then click a picture of that picture with resolution X, you will end up with a picture of resolution X. The same is true if the order is reversed as well.

We have for $i, j, \in [n],\ i < j$,

$$\begin{aligned}\mathbb{E}\left[\Delta_i\Delta_j\right] &= \mathbb{E}\Big[\big(Z_i - Z_{i-1}\big)\big(Z_j - Z_{j-1}\big)\Big] \\ &= \mathbb{E}\Big[\mathbb{E}\left[\big(Z_i - Z_{i-1}\big)\big(Z_j - Z_{j-1}\big)\middle|X^i\right]\Big] \\ &= \mathbb{E}\Big[\big(Z_i - Z_{i-1}\big)\mathbb{E}\left[Z_j - Z_{j-1}\middle|X^i\right]\Big] \\ &= 0\end{aligned}$$

where in the second step we used the law of iterated expectations and the last step follows by (2.4). Since $\mathbb{E}\Delta_i = 0$ for all $i \in [n]$, we have

$$\mathbb{E}\Delta_i \cdot \mathbb{E}\Delta_j = 0 = \mathbb{E}[\Delta_i\Delta_j]$$

which completes the proof. □

Remark: **A stronger version of McDiarmid's inequality** (which we will not prove here) exists – with 4 times the exponent that we have obtained.

Example 2.5. *(Longest Common Subsequence)* A version of the longest common subsequence (LCS) problem is as follows: let $X_1, \ldots, X_n$ and $Y_1, \ldots, Y_n$ be two sequences of fair coin flips. Define Z as the length of the longest subsequence that appears in both sequences, that is,

$$Z = \max\{k : X_{i_1} = Y_{j_1}, \ldots, X_{i_k} = Y_{j_k} \text{ where } 1 \le i_1 \le \cdots \le i_k \le n \text{ and } 1 \le j_1 \le \cdots \le j_k \le n\}$$

It is known that $\mathbb{E}Z/n$ converges to some $\gamma \in (0.75796, 0.83763)$ whose value is *conjectured* to be $2(\sqrt{2} - 1)$.

Our interest is in understanding if the mean is a good proxy for the (random) LCS length. Observe that changing one coin flip cannot change the length of the longest common subsequence by more than one, so Z satisfies the BDP with $c_i = 1$. Thus, by McDiarmid's inequality, Z is strongly (exponentially) concentrated around its mean which allows us to conclude that $\mathbb{E}Z$ faithfully represents the (random) LCS length.

Chapter 3

Measures of randomness

Signal processing, machine learning, and statistics, in simplistic terms, may be thought of as formal ways of extracting useful "information" from signals/data. To evaluate how efficient an algorithm is or to design "principles of processing information" in all these fields, it is imperative to have appropriate measures of randomness or information. Information theory provides the foundations necessary for this. To begin with, we define desirable axioms that a function that measures the information content in a probabilistic source must satisfy and show that the resulting function is of a certain form, referred to as the *Shannon entropy* of the source.

3.1 Shannon entropy

Let $P = (p_1, \ldots, p_k)$ be a PMF on an alphabet of size $k \geq 2$. A function f that "measures the information" in P must satisfy the following axioms:

1. f is continuous in $p_i, i \in [k]$.

2. If all p_i are equal, i.e., $p_i = 1/k$, then f is a monotonically increasing function of k (for ease of scaling, we assume that the smallest among these, namely, $f(1/2, 1/2) = 1$).

3. If a choice is broken down into two successive choices, the original f is the weighted sum of individual values of f.

   See Fig. 3.1; on the left, we have three possibilities, $p_1 = 1/2, p_2 = 1/3, p_3 = 1/6$. On the right, we first choose between two possibilities each with probability $1/2$, and if the second occurs make another choice with probabilities $2/3, 1/3$. The final results have the same probabilities as before. We require, in this special case, that
   $$f(1/2, 1/3, 1/6) = f(1/2, 1/2) + \frac{1}{2} f(2/3, 1/3).$$
   The coefficient $1/2$ is because this second choice only occurs half the time.

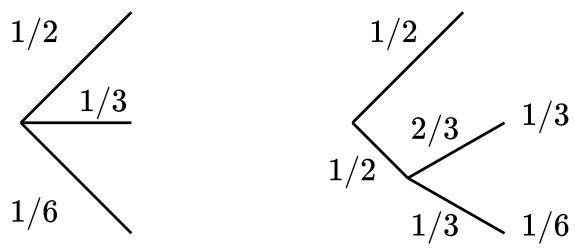


Figure 3.1: Decomposition of a choice from three possibilities

We show that the only function satisfying these axioms is the **Shannon entropy** function defined by

$$H(P) = -\sum_{i=1}^{k} p_i \log_2 p_i. \tag{3.1}$$

with the convention that $0 \log_2 0 = 0$.

Exercise 3.1: Reading exercise

Check out the original paper by Claude E. Shannon, "A mathematical theory of communication" wherein the idea of Shannon entropy was first presented. This profound paper laid the foundations for the information age by developing a mathematical theory of information, compression, and communication.

Theorem 3.1. *The only function satisfying the axioms above is given by* (3.1).

Proof. Let f be the function that satisfies all the axioms and let

$$u(k) := f\left(\frac{1}{k}, \ldots, \frac{1}{k}\right),$$

the value of f for the uniform distribution on an alphabet of size k. For positive integers m and n, by the third axiom, we have

$$\begin{aligned} u(mn) &= f\left(\frac{1}{mn}, \ldots, \frac{1}{mn}\right) \\ &= f\left(\frac{1}{m}, \ldots, \frac{1}{m}\right) + \sum_{j=1}^{m} \frac{1}{m} \cdot f\left(\frac{1}{n}, \ldots, \frac{1}{n}\right) \\ &= f\left(\frac{1}{m}, \ldots, \frac{1}{m}\right) + f\left(\frac{1}{n}, \ldots, \frac{1}{n}\right) \\ &= u(m) + u(n). \end{aligned}$$

Therefore,

$$u\left(2^{x+y}\right) = u\left(2^x \cdot 2^y\right) = u\left(2^x\right) + u\left(2^y\right).$$

Denoting by g, the function $(u \circ \exp_2)$, we have

$$g(x+y) = g(x) + g(y).$$

Thus, g is linear and hence

$$g(x) = cx$$

for some constant c. Then, for $x = \log_2 y$, we have

$$u(y) = u(2^{\log_2 y}) = (u \circ \exp_2)(\log_2 y) = g(\log_2 y) = c \log_2 y.$$

Since $u(2) = 1$ (by axiom 2), we have $c = 1$ and hence

$$u(m) = \log_2 m.$$

Let m and n be positive integers such that $m < n$. Again, by the third axiom, we have

$$\begin{aligned} f\left(\frac{1}{n}, \ldots, \frac{1}{n}\right) &= f\left(\frac{m}{n}, \frac{n-m}{n}\right) + \frac{m}{n} f\left(\frac{1}{m}, \ldots, \frac{1}{m}\right) + \left(\frac{n-m}{n}\right) f\left(\frac{1}{n-m}, \ldots, \frac{1}{n-m}\right) \\ &= f\left(\frac{m}{n}, \frac{n-m}{n}\right) + \frac{m}{n} \cdot u(m) + \frac{n-m}{n} \cdot u(n-m) \\ &= f\left(\frac{m}{n}, \frac{n-m}{n}\right) + \frac{m}{n} \log_2 m + \frac{n-m}{n} \log_2(n-m), \end{aligned}$$

rearranging which yields

$$\begin{aligned} f\left(\frac{m}{n}, 1-\frac{m}{n}\right) &= \log_2 n - \frac{m}{n}\log_2 m - \left(1-\frac{m}{n}\right)\log_2(n-m) \\ &= \frac{m}{n}\log_2 n + \left(1-\frac{m}{n}\right)\log_2 n - \frac{m}{n}\log_2 m - \left(1-\frac{m}{n}\right)\log_2(n-m) \\ &= -\frac{m}{n}\log_2\left(\frac{m}{n}\right) - \left(1-\frac{m}{n}\right)\log_2\left(1-\frac{m}{n}\right). \end{aligned}$$

By the continuity axiom (axiom 1), we have for any $p \in [0,1]$,

$$f(p, 1-p) = -p\log_2 p - (1-p)\log_2(1-p).$$

The proof is completed via mathematical induction on the alphabet size and is left as an exercise. □

Exercise 3.1: Mathematical induction on k

Complete the proof of Theorem 3.1.
Hint: Assume true for $k-1$. Write

$$f(p_1, \ldots, p_k) = f(1-p_k, p_k) + (1-p_k) f\left(\frac{p_1}{1-p_k}, \ldots, \frac{p_{k-1}}{1-p_k}\right).$$

Remark: Recall that we fixed the value of the constant $c = 1$ using the requirement in the second axiom, namely, $H(1/2, 1/2) = 1$. If we remove this requirement, changing the value of c is equivalent to keeping $c = 1$ and changing the base of the logarithm in (3.1) appropriately; for bases $2, e$, and 10, the units of $H(P)$ are *bits*, *nats* and *hartleys*, respectively.
Notation: In the sequel, unless specified otherwise, log stands for $\log_2$ and the unit of H is bits. We also use $H(X)$ to denote $H(P)$ when $X \sim P$.

3.1.1 Properties of Shannon entropy

Definition 3.1. For (X, Y) with a joint distribution $P_{XY} = P_X W_{Y|X}$, that is, for every $(x, y) \in \mathcal{X} \times \mathcal{Y}$,

$$P_{XY}(x, y) = P_X(x) W_{Y|X}(y|x),$$

let

$$H(X, Y) := -\sum_{x \in \mathcal{X}, y \in \mathcal{Y}} P_{XY}(x, y) \log P_{XY}(x, y).$$

Here, $W_{Y|X}(\cdot|\cdot)$ is referred to as the **channel** from X to Y and $H(X, Y)$ is called the **joint entropy** of (X, Y).

Exercise 3.2: Properties of H

1. Let $X \sim P$. Consider the random variable $Z = -\log_2 P(X)$. What is $\mathbb{E}Z$?
2. Show that for any random variable X, $H(X) \geq 0$ with equality iff X is a constant.
   Hint: $\mathbb{P}\{X \geq 0\} = 1 \implies \mathbb{E}X \geq 0$.

Definition 3.2. Let $H(Y|X = x)$ denote the entropy of the conditional distribution of Y given $X = x$. That is,

$$\begin{aligned} H(Y|X = x) &= -\sum_y \mathbb{P}[Y = y|X = x] \log \mathbb{P}[Y = y|X = x] \\ &= -\sum_y W_{Y|X}(y|x) \log W_{Y|X}(y|x). \end{aligned}$$

The quantity

$$\mathbb{E}_X\left[H(Y|X=x)\right] = -\sum_x \sum_y P_X(x) W_{Y|X}(y|x) \log W_{Y|X}(y|x) = -\sum_{x,y} P_{XY}(x,y) \log W_{Y|X}(y|x).$$

is called the **conditional entropy of Y given X** and is denoted $H(Y|X)$. We can interpret $H(X)$ as a measure of uncertainty in X and $H(Y|X)$ as the uncertainty in Y once X is revealed. Thus, it is natural that the uncertainty in the pair (X,Y) is

$$H(X,Y) = H(X) + H(Y|X).$$

Exercise 3.3: Properties of H (continued)

1. Prove: $H(X,Y) = H(X) + H(Y|X)$.
2. Let $\mathcal{X} = \mathcal{Y} = \{0,1\}$. Consider the following channel:

$$W_{Y|X}(0|1) = W_{Y|X}(1|0) = \alpha.$$

   (a) Calculate $W_{Y|X}(0|0)$ and $W_{Y|X}(1|1)$.
   (b) Let $X \sim \mathrm{Ber}(p)$. Calculate $H(X), H(Y|X)$, and $H(X,Y)$.
3. Let $f : \mathcal{X} \to \mathbb{R}$ be a deterministic function. Show that

$$H(X) \geq H(f(X)).$$

   When does equality hold?
   Hint: Expand $H(X, f(X))$ as $H(X) + H(f(X)|X)$ and $H(f(X)) + H(X|f(X))$ and compare.
4. Prove the *log-sum inequality*: Suppose $a_i, b_i \geq 0, 1 \leq i \leq k$. Then,

$$\sum_{i=1}^{k} \left(a_i \log \frac{a_i}{b_i} \right) \geq \left(\sum_{i=1}^{k} a_i \right) \log \frac{\sum_{i=1}^{k} a_i}{\sum_{i=1}^{k} b_i}$$

   with the convention $0 \log \frac{0}{0} = 0$. WLOG, we can assume $\sum_{i=1}^{k} a_i = \sum_{i=1}^{k} b_i = 1$ (Why?).
   Hint: Use the inequality $\ln x \leq x - 1$.
5. Using the log-sum inequality, verify that

   (a) $H(X) \leq \log_2 |\mathcal{X}|$.
   (b) $H(Y) \geq H(Y|X)$.

   Hint: For the latter, write

$$\begin{aligned} H(Y) &= \sum_y P_Y(y) \log \frac{1}{P_Y(y)} \\ &= \sum_y \left[\left(\sum_x P_X(x) W_{Y|X}(y|x) \right) \log \frac{\sum_x P_X(x)}{\sum_x P_X(x) W_{Y|X}(y|x)} \right] \end{aligned}$$

   and let i range over $x \in \mathcal{X}$, $a_i = P_X(x) W_{Y|X}(y|x)$ and $b_i = P_X(x)$.
6. Is $H(X,Y) \leq H(X) + H(Y)$? Justify.

Definition 3.3. A set $\mathcal{C} \subset \mathbb{R}$ is a **convex set** if for all $x, y \in \mathcal{C}$, for every $\theta \in [0,1]$,

$$\theta x + (1-\theta)y \in \mathcal{C}.$$

That is, if two points belong to $\mathcal{C}$, then all the points on the straight line joining the two points also belong to $\mathcal{C}$.

Definition 3.4. Let $\mathcal{C}$ be a convex set. Then, a function $f : \mathcal{C} \to \mathbb{R}$ is said to be **convex** if for every $\theta \in [0,1]$ and for every x, y in $\mathcal{C}$, we have

$$f(\theta x + (1-\theta)y) \leq \theta f(x) + (1-\theta)f(y)$$

and is said to be **concave** if

$$f(\theta x + (1-\theta)y) \geq \theta f(x) + (1-\theta)f(y).$$

Note that f is convex $\iff$ $-f$ is concave.

Theorem 3.2. *(Jensen's inequality) For a convex function $f : \mathbb{R} \to \mathbb{R}$ and X a random variable,*

$$f(\mathbb{E}X) \leq \mathbb{E}[f(X)].$$

The proof follows from the definition of convex functions and mathematical induction (left as an exercise).

Exercise 3.4: Jensen's inequality

Prove Jensen's inequality.

Example 3.1. (Concavity of Shannon entropy) $H(P)$ is concave in P.

Proof. First, verify that the set of distributions is a convex set. That is, for $\theta \in [0,1]$ and P_0, P_1 discrete distributions on $\mathcal{X}$, check that

$$P_\theta = \theta P_0 + (1-\theta)P_1$$

contains non-negative entries that add up to 1. We want to show that

$$H(\theta P_0 + (1-\theta)P_1) \geq \theta H(P_0) + (1-\theta)H(P_1).$$

Let $X \sim \text{Ber}(1-\theta)$ and let

$$\begin{aligned} W_{Y|X}(y|0) &= P_0(y) \\ W_{Y|X}(y|1) &= P_1(y). \end{aligned}$$

Then,

$$\begin{aligned} H(Y|X) &= \sum_{x,y} P_X(x) H\left(W_{Y|X}(\cdot|x)\right) \\ &= P_X(0)H(P_0) + P_X(1)H(P_1) \\ &= \theta H(P_0) + (1-\theta)H(P_1). \end{aligned}$$

Now,

$$\begin{aligned} P_Y(y) &= \sum_x P_X(x) W_{Y|X}(y|x) \\ &= \theta P_0(y) + (1-\theta)P_1(y) \end{aligned}$$

whereby

$$H(Y) = H(\theta P_0 + (1-\theta)P_1).$$

The proof is complete by the fact $H(Y) \geq H(Y|X)$. □

3.1.2 Shannon's source coding theorem*

The quantity Shannon entropy arises in multiple contexts, the primary one being data compression. Suppose we have a source that emits n symbols i.i.d. P from an alphabet $\mathcal{X}$. What is the minimum number of bits with which we can store the sequence if we allow an error of ε in reconstruction? The answer is roughly $nH(P)$ for n large enough.

Definition 3.5. A **discrete source** X is a random variable with PMF P_X taking values in a finite set $\mathcal{X}$ (called the source alphabet).

Example 3.2. Consider a discrete source that emits 10-bit sequences with:

$$P_X(x) = \begin{cases} 0.495 & \text{if } x = 0000000000 \\ 0.495 & \text{if } x = 1111111111 \\ \frac{0.01}{1022} & \text{otherwise.} \end{cases}$$

– If we want to store and recover sequences making no mistake *with probability* 1, how many bits suffice to be allocated for storing a symbol emitted by the source? *Ans:* 10.
– If we want to store and recover sequences making no mistake *with probability* 0.99, how many bits suffice to be allocated for storing a symbol emitted by the source? *Ans:* 1. (Why?)

Thus, if we know the probability distribution of the source and if we allow for some error, it suffices to allocate fewer bits for the same source. This motivates the question: for a given probability distribution, is there a minimum number of bits with which one can guarantee the requirement on the error? We make the question more precise (and answer it in the affirmative) by considering the following model (see Figure 3.2):

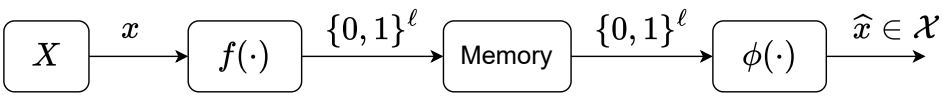


Figure 3.2: Compression schematic

Definition 3.6. A **(block) source code** (f, ϕ) of length ℓ comprises a *compression mapping*

$$f : \mathcal{X} \to \{0,1\}^\ell$$

and a *decompression mapping*

$$\phi : \{0,1\}^\ell \to \mathcal{X}.$$

Definition 3.7. The **probability of error** for (f, ϕ) is given by

$$e(f, \phi) := \mathbb{P}[\phi(f(X)) \neq X] = \sum_{x \in \mathcal{X}} P_X(x) \mathbf{1}_{\{\phi(f(x)) \neq x\}}.$$

Let

$$\mathcal{X}_0 = \{x \in \mathcal{X} : \phi(f(x)) = x\}$$

denote the set of symbols where we do not make a mistake. Therefore, ensuring that the probability of error does not exceed ε is equivalent to **designing** $\mathcal{X}_0$ such that

$$\sum_{x \in \mathcal{X}_0} P_X(x) = 1 - e(f, \phi) \geq 1 - \varepsilon$$

and choosing $\ell = \lceil \log |\mathcal{X}_0| \rceil$ so that we represent all the elements in $\mathcal{X}_0$ in a one-to-one fashion using ℓ-bit sequences. Now it remains to figure out how to build such a "large probability set" $\mathcal{X}_0$. We do this for a specific source – one that emits symbols i.i.d. from a known distribution.
Notation: We use P_{X^n} to denote the distribution of $X^n \sim$ i.i.d. P_X.

Definition 3.8. A **discrete memoryless source (DMS)** X^n of length n is a source with alphabet $\mathcal{X}^n = \mathcal{X} \times \cdots \times \mathcal{X}$, with sequences x^n as realizations, and with PMF

$$P_{X^n}(x^n) = \prod_{i=1}^{n} P_X(x_i).$$

Definition 3.9. (**Rate of a source code**) A source code of rate $R > 0$ with probability of error ε is a length nR source code for the DMS X^n with probability of error not exceeding ε.

Definition 3.10. A rate $R > 0$ is **ε-achievable** if there exists a source code of rate $R > 0$ with probability of error ε for all n sufficiently large.

Definition 3.11. (**Optimal rates**) The smallest among all ε-achievable rates is denoted R^*_ε and let

$$R^* = \lim_{\varepsilon \to 0} R^*_\varepsilon.$$

With these definitions in hand, we seek to build a "large probability set" for the DMS X^n and estimate its size (for large n) which leads to a characterization of the optimal rate for the source code of X^n. For a fixed $\delta > 0$, consider the set of sequences in $\mathcal{X}^n$,

$$A^n_\delta := \left\{ x^n \in \mathcal{X}^n : \left| -\frac{1}{n} \sum_{i=1}^{n} \log P_X(x_i) - H(P_X) \right| \leq \delta \right\}$$

The set A^n_δ is called the **δ-typical set** (or simply, typical set). Observe that

$$\mathbb{E}\left[-\frac{1}{n} \sum_{i=1}^{n} \log P_X(x_i) \right] = H(P_X)$$

and hence by Chebyshev's inequality (or WLLN), we have the following lemma:

Lemma 3.1 (AEP)**.** *For every $\delta > 0$, we have*

$$\lim_{n \to \infty} P_{X^n}(A^n_\delta) = 1.$$

That is, for every $\varepsilon > 0$, for n sufficiently large,

$$P_{X^n}(A^n_\delta) \geq 1 - \varepsilon.$$

Observe that for all the sequences $x^n \in A^n_\delta$, we have

$$2^{-n(H(P_X)+\delta)} \leq P_{X^n}(x^n) \leq 2^{-n(H(P_X)-\delta)}. \tag{3.2}$$

That is, for n large enough, the set has probability 1 and every sequence in the set has "more or less" the same probability. This is known as **asymptotic equipartition property** (AEP). By Lemma 3.1, the typical set is a good candidate for the "large probability set" we seek. We estimate the length ℓ of the source code for encoding the sequences in A^n_δ in a one-to-one fashion into ℓ-bit sequences. In other words, we estimate the size of the typical set.

Lemma 3.2. *For n sufficiently large, we have*

$$(1 - \delta) 2^{n(H(P_X)-\delta)} \leq |A^n_\delta| \leq 2^{n(H(P_X)+\delta)}.$$

That is,

$$\frac{1}{n} \log |A^n_\delta| \approx H(P_X)$$

for all n sufficiently large.

Proof. We have, by (3.2),

$$1 \geq P_{X^n}(A_\delta^n) = \sum_{x^n \in A_\delta^n} P_{X^n}(x^n) \geq \sum_{x^n \in A_\delta^n} 2^{-n(H(P_X)+\delta)} = |A_\delta^n| \, 2^{-n(H(P_X)+\delta)}$$

which proves the upper bound. For n sufficiently large, by Lemma 3.1, we have, $P_{X^n}(A_\delta^n) \geq 1 - \delta$ and hence by (3.2), we have

$$1 - \delta \leq P_{X^n}(A_\delta^n) = \sum_{x^n \in A_\delta^n} P_{X^n}(x^n) \leq \sum_{x^n \in A_\delta^n} 2^{-n(H(P_X)-\delta)} = |A_\delta^n| \, 2^{-n(H(P_X)-\delta)}$$

which proves the lower bound. □

The set A_δ^n is a "large probability set" and we showed in Lemma 3.2 that the size of A_δ^n is roughly $2^{nH(P_X)}$. In fact, this is true for any large probability set comprising sequences in $\mathcal{X}^n$.

Lemma 3.3. *For $0 < \varepsilon < 1$, let B^n be a set of sequences in $\mathcal{X}^n$ such that*

$$P_{X^n}(B^n) \geq 1 - \varepsilon.$$

Then, for every $\delta > 0$ and n sufficiently large,

$$\frac{1}{n} \log_2 |B^n| \geq H(P_X) - \delta'.$$

where $\delta' > \delta$.

Proof. By Lemma 3.1, for n sufficiently large, we have

$$P_{X^n}(A_\delta^n) \geq 1 - \varepsilon'.$$

Then,

$$\begin{aligned} P_{X^n}(A_\delta^n \cap B^n) &= P_{X^n}(A_\delta^n) + P_{X^n}(B^n) - P_{X^n}(A_\delta^n \cup B^n) \\ &\geq 1 - \varepsilon' + 1 - \varepsilon - 1 \\ &= 1 - \varepsilon - \varepsilon'. \end{aligned}$$

Using this, we have

$$\begin{aligned} 1 - \varepsilon - \varepsilon' &\leq \sum_{x^n \in A_\delta^n \cap B^n} P_{X^n}(x^n) \\ &\leq \sum_{x^n \in A_\delta^n \cap B^n} 2^{-n(H(P_X)-\delta)} \\ &= |A_\delta^n \cap B^n| \cdot 2^{-n(H(P_X)-\delta)} \\ &\leq |B^n| \cdot 2^{-n(H(P_X)-\delta)} \end{aligned}$$

where the second inequality follows from (3.2). □

We are ready to prove the main theorem of this section.

Theorem 3.3 (Shannon's source coding theorem)**.** *For a DMS with common distribution P_X, for all $0 < \varepsilon < 1$,*

$$R_\varepsilon^* = R^* = H(P_X).$$

Proof. **(Achievability)** Since for n sufficiently large, we have

$$P_{X^n}(A_\delta^n) \geq 1 - \varepsilon,$$

$\frac{\lceil \log_2 |A_\delta^n| \rceil}{n}$ is an *achievable rate* for the DMS X^n (see Definition 3.9). Hence, by Lemma 3.2, we have

$$\begin{aligned} R_\varepsilon^* &\leq \frac{\log_2 |A_\delta^n| + 1}{n} \\ &\leq \frac{n(H(P_X) + \delta) + 1}{n} \\ &\leq H(P_X) + \delta' \end{aligned}$$

for n sufficiently large where $\delta' > \delta$. Since this is true for every $\delta > 0$, it follows that

$$R_\varepsilon^* \leq H(P_X).$$

(Converse) Let (f^*, ϕ^*) be the source code with probability of error ε and the smallest rate. Then,

$$C^n := \{x^n \in \mathcal{X}^n : \phi^*(f^*(x^n)) = x^n\}$$

must satisfy

$$P_{X^n}(C^n) \geq 1 - \varepsilon.$$

Since Lemma 3.3 is true for set of sequences in $\mathcal{X}^n$ with probability at least $1 - \varepsilon$, we have, for n sufficiently large,

$$\begin{aligned} R_\varepsilon^* &= \frac{\log_2 |C^n|}{n} \\ &\geq H(P_X) - \delta' \end{aligned}$$

where $\delta' > \delta$. Since this is true for every $\delta > 0$, it follows that

$$R_\varepsilon^* \geq H(P_X).$$

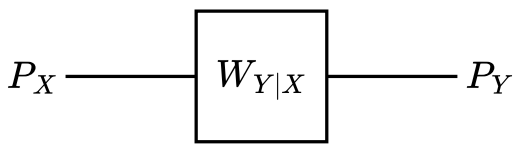
The proof is complete by observing that

$$R^* = \lim_{\varepsilon \to 0} R_\epsilon^* = H(P_X).$$

□

3.2 Data processing inequalities

Statistical inference entails processing data. Intuitively, it is easier to distinguish between two distributions using data generated from either of the distributions when the data is clean/complete as opposed to when the data is noisy/partial. Processing data would potentially *shrink the distance between the distributions* and make distinction harder. This idea is formalized by what are known as **data processing inequalities** (DPI). WLOG, data processing may be thought of as passing the data through a channel (see Figure).



DPI simply states that for most of the *distance measures* between distributions (some of which we discuss in the sequel), our intuition that "whatever you can distinguish using processed data, you can do using raw data" is correct. We shall prove a version of DPI for each distance measure.

3.3 Total variation distance

A canonical task that underlies statistical learning is *hypothesis testing*. Suppose we observe a sample $X \in \mathcal{X}$. How difficult it is to determine if X was generated from the distribution P or the distribution Q? Intuitively, the "farther" P and Q are, the easier this task is. This entails defining an appropriate distance metric on the space of probability distributions on the given alphabet $\mathcal{X}$. Here since there are only two hypotheses to distinguish between, this problem is called binary hypothesis testing. In the context of binary hypothesis testing, the appropriate metric is the *total variation distance.*

Definition 3.12. The **total variation distance** between P and Q on $\mathcal{X}$ is defined as

$$d_{\mathrm{TV}}(P, Q) := \frac{1}{2} \sum_{x} |P(x) - Q(x)|.$$

Exercise 3.5: Properties of total variation distance

Prove the following:

i. *Nonnegativity*
$$d_{\mathrm{TV}}(P, Q) \geq 0 \qquad \text{with equality iff } P = Q.$$

ii. *Symmetry*
$$d_{\mathrm{TV}}(P, Q) = d_{\mathrm{TV}}(Q, P).$$

iii. *Triangle inequality*
$$d_{\mathrm{TV}}(P, Q) \leq d_{\mathrm{TV}}(P, R) + d_{\mathrm{TV}}(R, Q).$$

iv. *Upper bound of* 1
$$d_{\mathrm{TV}}(P, Q) \leq 1.$$

3.3.1 Bayesian hypothesis testing

In the binary hypothesis testing problem, formally,

$$\mathcal{H}_0 : X \sim P$$
$$\mathcal{H}_1 : X \sim Q$$

where $\mathcal{H}_0$ is referred to as the *null hypothesis* and $\mathcal{H}_0$ is referred to as the *alternate hypothesis*. let $T : \mathcal{X} \to \{P, Q\}$ denote the decision (also referred to as the test). We allow T to be a channel and let $T(P|x) = 1 - T(Q|x)$ denote the probability that the test declares P for input $x \in \mathcal{X}$.

If $X \sim P$, then T makes an error if the output is Q which is called a *false alarm* (type I error) and it occurs with probability

$$P_{\mathrm{FA}}(T) := \sum_{x \in \mathcal{X}} P(x) T(Q|x).$$

If $X \sim Q$, then T makes an error if the output is P which is called a *missed detection* (type II error) and it occurs with probability

$$P_{\mathrm{MD}}(T) := \sum_{x \in \mathcal{X}} Q(x) T(P|x).$$

Further, in a Bayesian framework, one can assign priors, say $\theta \in (0, 1)$ to the probability that the true hypothesis is P. Under this setting, the average probability of error is defined as

$$P_e^{\theta}(T|P, Q) := \theta \cdot P_{\mathrm{FA}}(T) + (1 - \theta) \cdot P_{\mathrm{MD}}(T).$$

We restrict to $\theta = 1/2$ and drop θ from the notation to yield

$$P_e(T|P,Q) = \frac{1}{2}\left\{\sum_{x\in\mathcal{X}} P(x)T(Q|x) + \sum_{x\in\mathcal{X}} Q(x)T(P|x)\right\}$$

The minimum Bayes' average probability of error is the minimum of $P_e(T|P,Q)$ over all channels $T : \mathcal{X} \to \{P,Q\}$ and can be treated as a measure of difficulty of the hypothesis testing problem itself. Formally, it is defined as

$$P_e(P,Q) := \min_T P_e(T|P,Q).$$

The next result relates $P_e(P,Q)$ to $d_{\mathrm{TV}}(P,Q)$ via an identity.

Theorem 3.4 (Bayes' test)**.** *For distributions P and Q on $\mathcal{X}$, $P_e(P,Q)$ is attained by the deterministic test that outputs P for $x \in \mathcal{X}$ whenever $P(x) \geq Q(x)$ and outputs Q otherwise. That is,*

$$T^*(P|x) = \begin{cases} 1 & \text{if } P(x) \geq Q(x) \\ 0 & \text{otherwise.} \end{cases}$$

Furthermore, the minimum average probability of error is given by

$$P_e(P,Q) = \frac{1}{2}\big[1 - d_{\mathrm{TV}}(P,Q)\big].$$

Proof. Indeed, the average probability of error for T^* is given by

$$\begin{aligned}
P_e(T^*|P,Q) &= \frac{1}{2}\left[\sum_{x\in\mathcal{X}} P(x)T^*(Q|x) + \sum_{x\in\mathcal{X}} Q(x)T^*(P|x)\right] \\
&= \frac{1}{2}\left[\sum_{x\in\mathcal{X}} P(x)\mathbf{1}_{\{P(x)<Q(x)\}} + \sum_{x\in\mathcal{X}} Q(x)\mathbf{1}_{\{P(x)\geq Q(x)\}}\right] \\
&= \frac{1}{2}\left[\sum_{x\in\mathcal{X}} P(x)\left[1 - \mathbf{1}_{\{P(x)\geq Q(x)\}}\right] + \sum_{x\in\mathcal{X}} Q(x)\mathbf{1}_{\{P(x)\geq Q(x)\}}\right] \\
&= \frac{1}{2}\left[\sum_{x\in\mathcal{X}} P(x) - \sum_{x\in\mathcal{X}} [P(x) - Q(x)]\,\mathbf{1}_{\{P(x)\geq Q(x)\}}\right] \\
&= \frac{1}{2}\left[1 - \sum_{x\in\mathcal{X}:P(x)\geq Q(x)} [P(x) - Q(x)]\right] \\
&= \frac{1}{2}\big[1 - d_{\mathrm{TV}}(P,Q)\big].
\end{aligned}$$

where the last step is left as an exercise.

Exercise 3.6: Equivalent definition of total variation distance

Show that

$$d_{\mathrm{TV}}(P,Q) = \sum_{x\in\mathcal{X}:P(x)\geq Q(x)} P(x) - Q(x).$$

It remains to show that for every T,

$$P_e(T|P,Q) \geq P_e(T^*|P,Q).$$

We have

$$\begin{aligned}
2P_e(T|P,Q) &= \sum_{x\in\mathcal{X}} P(x)T(Q|x) + \sum_{x\in\mathcal{X}} Q(x)T(P|x) \\
&= \sum_{x\in\mathcal{X}} [P(x)-Q(x)]\,T(P|x) + \sum_{x\in\mathcal{X}} Q(x) \\
&\geq \sum_{x\in\mathcal{X}:P(x)<Q(x)} [P(x)-Q(x)]\,T(P|x) + \sum_{x\in\mathcal{X}} Q(x) \\
&\geq \sum_{x\in\mathcal{X}:P(x)<Q(x)} [P(x)-Q(x)] + \sum_{x\in\mathcal{X}} Q(x) \\
&= \sum_{x\in\mathcal{X}:P(x)<Q(x)} [P(x)-Q(x)]\,T^*(Q|x) + \sum_{x\in\mathcal{X}} Q(x) \\
&= 2P_e(T^*|P,Q).
\end{aligned}$$

□

Observe that the theorem states that the optimal test T^* is deterministic. A deterministic test T is characterized by the *acceptance region*, namely the subset of $\mathcal{X}$, where the test declares P. That is,

$$\mathcal{A} := \{x \in \mathcal{X} : T(P|x) = 1\}.$$

The Bayesian average probability of error (with equal priors) is then given by

$$\begin{aligned}
P_e(T|P,Q) &= \frac{1}{2}\left[P(\mathcal{A}^c) + Q(\mathcal{A})\right] \\
&= \frac{1}{2}\left[1 - (P(\mathcal{A}) - Q(\mathcal{A}))\right].
\end{aligned}$$

Since (by Theorem 3.4) the optimal test is deterministic, the minimum average probability of error is given by

$$\begin{aligned}
P_e(P,Q) &= \min_{\mathcal{A}\subseteq\mathcal{X}} \frac{1}{2}\big[1 - (P(\mathcal{A}) - Q(\mathcal{A}))\big] \\
&= \frac{1}{2}\left[1 - \max_{\mathcal{A}\subseteq\mathcal{X}} (P(\mathcal{A}) - Q(\mathcal{A}))\right]
\end{aligned}$$

from which it follows that indeed

$$d_{\mathrm{TV}}(P,Q) = \max_{\mathcal{A}\subseteq\mathcal{X}} P(\mathcal{A}) - Q(\mathcal{A}).$$

Exercise 3.7: Equivalent definition of total variation distance

Show using the definition that

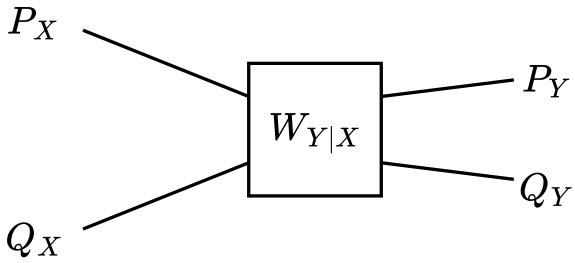
$$d_{\mathrm{TV}}(P,Q) = \max_{\mathcal{A}\subseteq\mathcal{X}} P(\mathcal{A}) - Q(\mathcal{A}).$$

Next, we exhibit the **DPI for $d_{\mathrm{TV}}(P,Q)$ and $P_e(P,Q)$**. Suppose two distributions P_X and Q_X are passed through the same channel $W_{Y|X}$ to yield P_Y and Q_Y, respectively (see Figure). That is,

$$\begin{aligned}
P_Y &= W_{Y|X}P_X \\
Q_Y &= W_{Y|X}Q_X.
\end{aligned}$$

Then,

$$d_{\mathrm{TV}}(P_Y,Q_Y) \leq d_{\mathrm{TV}}(P_X,Q_X).$$



Proof. We have

$$\begin{aligned}
2d_{\mathrm{TV}}(P_Y, Q_Y) &= \sum_y |P_Y(y) - Q_Y(y)| \\
&= \sum_y \Big| \sum_x \left[P_X(x)W_{Y|X}(y|x) - Q_X(x)W_{Y|X}(y|x)\right] \Big| \\
&= \sum_y \Big| \sum_x W_{Y|X}(y|x) \left[P_X(x) - Q_X(x)\right] \Big| \\
&\leq \sum_y \sum_x W_{Y|X}(y|x) \Big| P_X(x) - Q_X(x) \Big| \\
&= \sum_x \sum_y W_{Y|X}(y|x) \Big| P_X(x) - Q_X(x) \Big| \\
&= \sum_x |P_X(x) - Q_X(x)| \\
&= 2d_{\mathrm{TV}}(P_X, Q_X)
\end{aligned}$$

where the inequality is by triangle inequality. □

Recall that the minimum Bayes' average probability of error for a binary hypothesis testing problem is the minimum average (with equal priors) probability of error over all tests for the problem. Since the optimal test for P_Y versus Q_Y can be used as a test for P_X versus Q_X by transforming X into Y using $W_{Y|X}$, we have

$$P_e(P_X, Q_X) \leq P_e(P_Y, Q_Y).$$

Observe that this is equivalent to the DPI for total variation distance.

3.4 Kullback-Leibler divergence

In the Bayesian framework, for a given test T, we sought to minimize a convex combination of $P_{\mathrm{FA}}(T)$ and $P_{\mathrm{MD}}(T)$ and in the particular case of uniform priors showed that the minimum Bayes' average probability of error is characterized by $d_{\mathrm{TV}}(P, Q)$. Alternatively, one can seek to minimize $P_{\mathrm{MD}}(T)$ subject to the constraint that $P_{\mathrm{FA}}(T) \leq \varepsilon$ where $0 < \varepsilon < 1$ is chosen by us. Then, the quantity corresponding to $P_e(P, Q)$ is given by

$$\beta_\varepsilon(P, Q) := \min_{T : P_{\mathrm{FA}}(T) \leq \varepsilon} P_{\mathrm{MD}}(T)$$

and could be thought of as a "measure of distance" between P and Q. It captures the fundamental tradeoff between errors of type I and II. Just like $P_e(P, Q)$ is related to $d_{\mathrm{TV}}(P, Q)$, under this setting, $\beta_\varepsilon(P, Q)$ is related to a quantity called *Kullback-Leibler (KL) divergence* (also referred to as *relative entropy*). Although not a metric, KL divergence "measures the distance" between probability distributions.

Definition 3.13. The **KL divergence** between discrete distributions P and Q on $\mathcal{X}$ is defined as

$$D(P\|Q) = \begin{cases} \sum_{x\in\mathcal{X}} P(x)\log\frac{P(x)}{Q(x)} & \text{if } P(x)=0 \text{ whenever } Q(x)=0 \\ +\infty & \text{otherwise} \end{cases}$$

with the convention $0\log\frac{0}{0}=0$.

Note that

$$D(P\|Q) = -\sum_{x\in\mathcal{X}} P(x)\log Q(x) - H(P)$$

where the first term on the right-side is often referred to as the *cross-entropy* between P and Q.

Exercise 3.8: KL divergence and Shannon entropy

Prove: For a distribution P on $\mathcal{X}$,

$$H(P) = \log|\mathcal{X}| - D(P\|P_U)$$

where P_U denotes the uniform distribution on $\mathcal{X}$.

Theorem 3.5 (Properties of KL divergence)**.** *For discrete distributions P and Q on $\mathcal{X}$, the following hold:*

1. *(Non-negativity) $D(P\|Q)\geq 0$ with equality if and only if $P=Q$.*
2. *(Convexity) $D(P\|Q)$ is a convex function of the pair (P,Q), where P and Q are distributions on $\mathcal{X}$.*

Proof. Non-negativity follows from log-sum inequality (see Exercise 3.3.4). For proving convexity, consider the pair of distributions (P_0,Q_0) and (P_1,Q_1) on $\mathcal{X}$ and for $\theta\in[0,1]$, let

$$\begin{aligned} P_\theta &= \theta P_0 + (1-\theta)P_1 \\ Q_\theta &= \theta Q_0 + (1-\theta)Q_1. \end{aligned}$$

Then, we have

$$D(P_\theta\|Q_\theta) = \sum_{x\in\mathcal{X}} \left[\theta P_0(x) + (1-\theta)P_1(x)\right]\log\frac{\theta P_0(x)+(1-\theta)P_1(x)}{\theta Q_0(x)+(1-\theta)Q_1(x)}.$$

For every $x\in\mathcal{X}$, applying log-sum inequality with $n=2$ and

$$\begin{aligned} a_1 &= \theta P_0(x); & a_2 &= (1-\theta)P_1(x), \\ b_1 &= \theta Q_0(x); & b_2 &= (1-\theta)Q_1(x), \end{aligned}$$

we obtain

$$(a_1+a_2)\log\frac{a_1+a_2}{b_1+b_2} \leq a_1\log\frac{a_1}{b_1} + a_2\log\frac{a_2}{b_2}.$$

That is,

$$\begin{aligned} D(P_\theta\|Q_\theta) &\leq \sum_{x\in\mathcal{X}}\left[\theta P_0(x)\log\frac{\theta P_0(x)}{\theta Q_0(x)} + (1-\theta)P_1(x)\log\frac{(1-\theta)P_1(x)}{(1-\theta)Q_1(x)}\right] \\ &= \theta D(P_0\|Q_0) + (1-\theta)D(P_1\|Q_1). \end{aligned}$$

□

Definition 3.14. For jointly distributed random variables (X,Y), the **conditional KL divergence** given P_X is defined as the expectation (w.r.t. P_X) of the KL divergence between the distributions $P_{Y|X=x}$ and $Q_{Y|X=x}$. That is,

$$D\left(P_{Y|X}\|Q_{Y|X}|P_X\right) := \mathbb{E}_{x\sim P_X} D\left(P_{Y|X=x}\|Q_{Y|X=x}\right)$$

Theorem 3.6 (Properties of KL divergence (contd.))**.** *For joint distributions P_{XY} and Q_{XY} on $\mathcal{X} \times \mathcal{Y}$, the following hold:*

1. (Monotonicity)
$$D(P_{XY} \| Q_{XY}) \geq D(P_Y \| Q_Y).$$

2. (Chain rule)
$$D\left(P_{X_1,X_2,\ldots,X_n} \| Q_{X_1,X_2,\ldots,X_n}\right) = \sum_{i=1}^{n} D\left(P_{X_i|X^{i-1}} \| Q_{X_i|X^{i-1}} | P_{X^{i-1}}\right).$$

3. (Tensorization)
$$D\left(\prod_{i=1}^{n} P_{X_i} \middle\| \prod_{i=1}^{n} Q_{X_i}\right) = \sum_{i=1}^{n} D\left(P_{X_i} \| Q_{X_i}\right).$$
In particular, (when X^n are i.i.d.)
$$D(P^n \| Q^n) = nD(P \| Q).$$

4. (Conditioning increases divergence) *Given $P_{Y|X}$ and $Q_{Y|X}$, let*
$$P_Y = P_X P_{Y|X}; \quad Q_Y = P_X Q_{Y|X}.$$
Then,
$$D(P_Y \| Q_Y) \leq D\left(P_{Y|X} \| Q_{Y|X} | P_X\right)$$
with equality if and only if $D\left(P_{X|Y} \| Q_{X|Y} | P_Y\right) = 0$.

Proof. We prove the chain rule for two random variables X and Y; the general result follows by applying the rule recursively. Tensorization follows from the chain rule and monotonicity follows from the chain rule for X and Y.

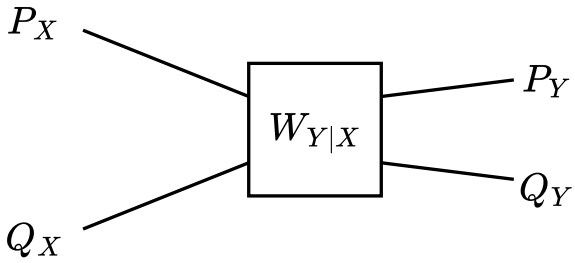
$$\begin{aligned}
D(P_{XY} \| Q_{XY}) &= \mathbb{E}\left[\log \frac{P_{XY}}{Q_{XY}}\right] \\
&= \mathbb{E}\left[\log \frac{P_X P_{Y|X}}{Q_X Q_{Y|X}}\right] \\
&= \mathbb{E}\left[\log \frac{P_X}{Q_X}\right] + \mathbb{E}\left[\log \frac{P_{Y|X}}{Q_{Y|X}}\right] \\
&= D(P_X \| Q_X) + D(P_{Y|X} \| Q_{Y|X} | P_X).
\end{aligned}$$

The property, conditioning increases divergence, is proved by applying the chain rule for X and Y in two different ways as follows:
$$D(P_{XY} \| Q_{XY}) = D(P_X \| Q_X) + D(P_{Y|X} \| Q_{Y|X} | P_X)$$
and
$$D(P_{XY} \| Q_{XY}) = D(P_Y \| Q_Y) + D(P_{X|Y} \| Q_{X|Y} | P_Y).$$
Since $P_X = Q_X$ and $D(P_{X|Y} \| Q_{X|Y} | P_Y) \geq 0$, the result follows. □

Next, we exhibit the **DPI for KL divergence**. Suppose two distributions P_X and Q_X are passed through the same channel $W_{Y|X}$ to yield P_Y and Q_Y, respectively (see Figure). That is,

$$\begin{aligned}
P_Y &= W_{Y|X} P_X \\
Q_Y &= W_{Y|X} Q_X.
\end{aligned}$$

Then,
$$D(P_Y \| Q_Y) \leq D(P_X \| Q_X).$$



Proof. For joint distributions $P_{XY} = P_X P_{Y|X}$ and $Q_{XY} = Q_X Q_{Y|X}$, by chain rule, we have

$$D(P_{XY}\|Q_{XY}) = D(P_X\|Q_X) + D(P_{Y|X}\|Q_{Y|X}|P_X)$$

and

$$D(P_{XY}\|Q_{XY}) = D(P_Y\|Q_Y) + D(P_{X|Y}\|Q_{X|Y}|P_Y).$$

Since, in our setting, there is only a single channel, we have $P_{Y|X} = Q_{Y|X} = W_{Y|X}$ and hence

$$D(P_{Y|X}\|Q_{Y|X}|P_X) = 0.$$

Observing that $D(P_{X|Y}\|Q_{X|Y}|P_Y) \geq 0$ completes the proof. □

3.4.1 Neyman-Pearson framework

Recall that we proposed as an alternative to the Bayesian framework, one wherein for $0 < \varepsilon < 1$, we seek tests T for which $P_{\text{FA}}(T) \leq \varepsilon$ and $P_{\text{MD}}(T)$ is minimized, the quantity of interest being

$$\beta_\varepsilon(P, Q) := \min_{T: P_{\text{FA}}(T) \leq \varepsilon} P_{\text{MD}}(T).$$

Before we characterize the relation between $\beta_\varepsilon(P, Q)$ and $D(P\|Q)$, we address the following natural question: what kind of tests achieve $\beta_\varepsilon(P, Q)$? The answer is given by Neyman-Pearson (NP) tests, also known as **likelihood ratio tests**, of the following form:

$$T_{\text{NP}}^{\gamma,\theta}(P|x) = \begin{cases} 1 & \text{if } \frac{P(x)}{Q(x)} > \gamma \\ \theta(x) & \text{if } \frac{P(x)}{Q(x)} = \gamma \\ 0 & \text{if } \frac{P(x)}{Q(x)} < \gamma. \end{cases}$$

Theorem 3.7 (NP lemma)**.** *(i)* ***Existence****: $\forall \varepsilon \in (0, 1)$, there exists a decision rule $T_{\text{NP}}^{\gamma,\theta}$ such that $\theta(x) = \theta$ for all $x \in \mathcal{X}$ and $P_{\text{FA}}(T_{\text{NP}}^{\gamma,\theta}) = \varepsilon$.*

(ii) ***Optimality****: Let T be any randomized decision rule with $P_{\text{FA}}(T) \leq \varepsilon$. Let $T_{\text{NP}}^{\gamma,\theta}$ be a randomized rule as defined above with $\gamma > 0$ an θ chosen such that $P_{\text{FA}}(T_{\text{NP}}^{\gamma,\theta}) = \varepsilon$. Then,*

$$P_{\text{MD}}(T_{\text{NP}}^{\gamma,\theta}) \leq P_{\text{MD}}(T).$$

Proof. We omit the proof of existence. We have, for $x \in \mathcal{X}$,

$$\left[T_{\text{NP}}^{\gamma,\theta}(P|x) - T(P|x)\right]\left[P(x) - \gamma Q(x)\right] \geq 0.$$

That is,

$$T_{\text{NP}}^{\gamma,\theta}(P|x)P(x) - T(P|x)P(x) \geq \gamma\left\{T_{\text{NP}}^{\gamma,\theta}(P|x)Q(x) - T(P|x)Q(x)\right\}.$$

Summing both sides over all $x \in \mathcal{X}$, we get

$$\sum_x T_{\mathrm{NP}}^{\gamma,\theta}(P|x)P(x) - \sum_x T(P|x)P(x) \geq \gamma \left\{ \sum_x T_{\mathrm{NP}}^{\gamma,\theta}(P|x)Q(x) - \sum_x T(P|x)Q(x) \right\}$$
$$(1 - P_{\mathrm{FA}}(T_{\mathrm{NP}}^{\gamma,\theta}) - (1 - P_{\mathrm{FA}}(T)) \geq \gamma \left\{ P_{\mathrm{MD}}(T_{\mathrm{NP}}^{\gamma,\theta}) - P_{\mathrm{MD}}(T) \right\}$$
$$P_{\mathrm{FA}}(T) - P_{\mathrm{FA}}(T_{\mathrm{NP}}^{\gamma,\theta}) \geq \gamma \left\{ P_{\mathrm{MD}}(T_{\mathrm{NP}}^{\gamma,\theta}) - P_{\mathrm{MD}}(T) \right\}$$

Since $P_{\mathrm{FA}}(T) \leq \varepsilon$ and $P_{\mathrm{FA}}(T_{\mathrm{NP}}^{\gamma,\theta}) = \varepsilon$, the left-side is less than or equal to zero. Since $\gamma > 0$, the proof of optimality follows. □

The import of the NP lemma is that it suffices to restrict attention to tests of the form $T_{\mathrm{NP}}^{\gamma,\theta}$. The next lemma relates $\beta_\varepsilon(P,Q)$ to $D(P\|Q)$; in fact, it is $-\log \beta_\varepsilon(P,Q)$ that is on the same scale as $D(P\|Q)$.

Lemma 3.4. *For a given $0 \leq \varepsilon < 1$, we have*

$$-\log \beta_\varepsilon(P,Q) \leq \frac{D(P\|Q) + h\left(\min\{\varepsilon, 1/2\}\right)}{1-\varepsilon}.$$

where $h(\cdot)$ denotes the binary entropy function.

Proof. For an arbitrary test T satisfying $P_{\mathrm{FA}}(T) \leq \varepsilon$, let Z be a random variable taking values in $\{P, Q\}$ and denoting the output of the test T. Then, under $\mathcal{H}_0$,

$$P_Z(z) := \sum_x P(x)T(z|x)$$

and under $\mathcal{H}_1$,

$$Q_Z(z) := \sum_x Q(x)T(z|x).$$

Then,

$$\begin{aligned} D(P_Z\|Q_Z) &= (1 - P_{\mathrm{FA}}(T)) \log \frac{1 - P_{\mathrm{FA}}(T)}{P_{\mathrm{MD}}(T)} + P_{\mathrm{FA}}(T) \log \frac{P_{\mathrm{FA}}(T)}{1 - P_{\mathrm{MD}}(T)} \\ &\geq (1 - P_{\mathrm{FA}}(T)) \log \frac{1}{P_{\mathrm{MD}}(T)} - h(P_{\mathrm{FA}}(T)) \\ &\geq (1-\varepsilon) \log \frac{1}{P_{\mathrm{MD}}(T)} - h\left(\min\{\varepsilon, 1/2\}\right). \end{aligned}$$

The above is true for all T satisfying $P_{\mathrm{FA}}(T) \leq \varepsilon$. Thus, the proof is complete by observing that

$$D(P_Z\|Q_Z) \leq D(P\|Q)$$

by DPI for KL divergence. □

Next, we exhibit the **DPI for** $\beta_\varepsilon(P,Q)$. Suppose two distributions P_X and Q_X are passed through the same channel $W_{Y|X}$ to yield P_Y and Q_Y, respectively. That is,

$$P_Y = W_{Y|X} P_X$$
$$Q_Y = W_{Y|X} Q_X.$$

Then,

$$\beta_\varepsilon(P_X, Q_X) \leq \beta_\varepsilon(P_Y, Q_Y).$$

Recall that in the Bayesian setting, we showed DPI for $P_e(P,Q)$ by arguing that the optimal test for P_Y versus Q_Y is a candidate test for P_X versus Q_X since we can transform X into Y by passing it through the channel $W_{Y|X}$. By contrast, here we need to ensure additionally that if the test $\widetilde{T}$ for P_Y versus Q_Y satisfies $P_{\mathrm{FA}}(\widetilde{T}) \leq \varepsilon$, then the composite test given by transforming X into Y and then applying $\widetilde{T}$ also satisfies the bound on the probability of false alarm. We use P and Q to denote the test outputs when the input is x as well as y.

Proof. For any test $\widetilde{T} : \mathcal{Y} \to \{P, Q\}$, the composite channel $W \circ \widetilde{T} : \mathcal{X} \to \{P, Q\}$ defined by

$$W \circ \widetilde{T}(z|x) := \sum_y W_{Y|X}(y|x)\widetilde{T}(z|y)$$

can be regarded as a test for $\mathcal{X}$. If $\widetilde{T}$ satisfies $P_{\mathrm{FA}}(\widetilde{T}) \leq \varepsilon$, we have

$$\begin{aligned} P_{\mathrm{FA}}(W \circ \widetilde{T}) &= \sum_x P_X(x) \sum_y W_{Y|X}(y|x)\widetilde{T}(Q|y) \\ &= \sum_y P_Y(y)\widetilde{T}(Q|y) \\ &\leq \varepsilon. \end{aligned}$$

Therefore,

$$\begin{aligned} \beta_\varepsilon(P, Q) &\leq P_{\mathrm{MD}}(W \circ \widetilde{T}) \\ &= \sum_x Q_X(x) \sum_y W_{Y|X}(y|x)\widetilde{T}(P|y) \\ &= \sum_y Q_Y(y)\widetilde{T}(P|y) \\ &= P_{\mathrm{MD}}(\widetilde{T}). \end{aligned}$$

Minimizing the right-side over all $\widetilde{T}$ satisfying $P_{\mathrm{FA}}(\widetilde{T}) \leq \varepsilon$ yields the result. □

3.4.2 Stein's lemma*

For $X_1, \ldots, X_n$ i.i.d. with

$$\begin{aligned} \mathcal{H}_0 &: X^n \sim P^n \\ \mathcal{H}_1 &: X^n \sim Q^n \end{aligned}$$

the quantity $\beta_\varepsilon(P^n, Q^n)$ decays exponentially in n. Stein's lemma states that for n large enough,

$$\beta_\varepsilon(P^n, Q^n) \approx 2^{-nD(P\|Q)}.$$

Definition 3.15. Given $0 < \varepsilon < 1$, an exponent E is **ε-achievable** if there exists (deterministic) tests with acceptance regions $\mathcal{A}_n$[1] such that

$$\begin{aligned} P^n(\mathcal{A}_n) &\geq 1 - \varepsilon, \\ Q^n(\mathcal{A}_n) &\leq 2^{-nE} \end{aligned}$$

for n large enough. Let E_ε^* denote the **largest ε-achievable** E. That is,

$$E_\varepsilon^* := \lim_{n\to\infty} \frac{-\log \beta_\varepsilon(P^n, Q^n)}{n}$$

and let

$$E^* = \lim_{\varepsilon\to 0} E_\varepsilon^*.$$

[1] Recall that a deterministic test is characterized by the acceptance region, namely the subset of $\mathcal{X}^n$ where the test declares $\mathcal{H}_0$.

For a fixed $\delta > 0$, consider the set of sequences in $\mathcal{X}^n$,

$$S_\delta^n := \left\{ x^n \in \mathcal{X}^n : \left| \frac{1}{n} \sum_{i=1}^n \log \frac{P(x_i)}{Q(x_i)} - D(P\|Q) \right| \leq \delta \right\}$$

Observe that

$$\mathbb{E}\left[\frac{1}{n} \sum_{i=1}^n \log \frac{P(x_i)}{Q(x_i)} \right] = D(P\|Q)$$

and hence by Chebyshev's inequality (or WLLN), we have the following lemma:

Lemma 3.5. *For every $\delta > 0$, we have*

$$\lim_{n\to\infty} P^n(S_\delta^n) = 1.$$

That is, for every $\varepsilon > 0$, for n sufficiently large,

$$P^n(S_\delta^n) \geq 1 - \varepsilon.$$

Observe that for all the sequences $x^n \in S_\delta^n$, we have

$$P^n(x^n) 2^{-n(D(P\|Q)+\delta)} \leq Q^n(x^n) \leq P^n(x^n) 2^{-n(D(P\|Q)-\delta)}. \tag{3.3}$$

Summing over all x^n in S_δ^n and applying lemma 3.5, we obtain

$$(1-\varepsilon) 2^{-n(D(P\|Q)+\delta)} \leq Q^n(S_\delta^n) \leq 2^{-n(D(P\|Q)-\delta)} \tag{3.4}$$

where the lower bound on $Q^n(S_\delta^n)$ holds for n large enough. In fact, a similar lower bound is true for any set of "large probability" sequences.

Lemma 3.6. *Let $T^n \subset \mathcal{X}^n$ be a set of sequences $x_1, \ldots, x_n$ such that $P^n(T^n) \geq 1 - \varepsilon$. Then,*

$$Q^n(T^n) \geq (1 - 2\varepsilon) 2^{-n(D(P\|Q)+\delta)}$$

Proof. By Lemma 3.5, for n sufficiently large, we have

$$P^n(S_\delta^n) \geq 1 - \varepsilon.$$

Then,

$$\begin{aligned} P^n(S_\delta^n \cap T^n) &= P^n(A_\delta^n) + P^n(T^n) - P^n(S_\delta^n \cup T^n) \\ &\geq 1 - \varepsilon + 1 - \varepsilon - 1 \\ &= 1 - 2\varepsilon. \end{aligned}$$

Using this, we have

$$\begin{aligned} Q^n(T^n) &\geq Q^n(S_\delta^n \cap T^n) \\ &= \sum_{x^n \in S_\delta^n \cap T^n} Q^n(x^n) \\ &\geq \sum_{x^n \in S_\delta^n \cap T^n} P^n(x^n) 2^{-n(D(P\|Q)+\delta)} \\ &= P^n(S_\delta^n \cap T^n) \cdot 2^{-n(D(P\|Q)+\delta)} \\ &\geq (1 - 2\varepsilon) 2^{-n(D(P\|Q)+\delta)} \end{aligned}$$

where the second inequality follows from (3.3). □

We are ready to state and prove Stein's lemma

Theorem 3.8 (Stein's lemma)**.** *For every* $0 < \varepsilon < 1$,

$$E^* = E^*_\varepsilon = D(P\|Q).$$

Proof. **(Achievability)** Consider the test with acceptance region S^n_δ. Then, for n large enough, by lemma 3.5,

$$P^n(S^n_\delta) \geq 1 - \varepsilon$$

and by (3.4),

$$-\log Q^n(S^n_\delta) \geq n\left(D(P\|Q) - \delta\right)$$

for every $\delta > 0$ whereby we have

$$\lim_{n\to\infty} \frac{-\log \beta_\varepsilon(P^n, Q^n)}{n} \geq D(P\|Q).$$

(Converse) For any test with acceptance region T^n satisfying $P^n(T^n) \geq 1 - \varepsilon$, by lemma 3.6, we have for n sufficiently large and $\delta > 0$,

$$-\frac{\log Q^n(T^n)}{n} \leq D(P\|Q) + \delta - \frac{\log(1-2\varepsilon)}{n}$$

whereby we have

$$\lim_{n\to\infty} \frac{-\log \beta_\varepsilon(P^n, Q^n)}{n} \leq D(P\|Q).$$

□

Exercise 3.9: Weak converse

Using lemma 3.4, show that $E^* \leq D(P\|Q)$.

3.4.3 Pinsker's inequality

Pinsker's inequality relates total variation distance and KL divergence. We use $D_2(p\|q)$ to denote the KL divergence between distributions $\mathrm{Ber}(p)$ and $\mathrm{Ber}(q)$. That is,

$$D_2(p\|q) := p\log_2 \frac{p}{q} + (1-p)\log_2 \frac{1-p}{1-q}.$$

Theorem 3.9 (Pinsker's inequality)**.** *For distributions P and Q on $\mathcal{X}$, we have*

$$D(P\|Q) \geq \frac{2}{\ln 2} d^2_{\mathrm{TV}}(P, Q)$$

where $D(P\|Q)$ is calculated using $\log_2$.

Proof. Let $A \subseteq \mathcal{X}$. Then, applying DPI for KL divergence to the channel $\mathbf{1}_A : \mathcal{X} \to \{0, 1\}$, we get

$$D(P\|Q) \geq D_2(P(A)\|Q(A)).$$

Assuming Pinsker's inequality for Bernoulli random variables (Exercise 3.10), we have

$$\sqrt{\frac{\ln 2}{2} D(P\|Q)} \geq |P(A) - Q(A)|.$$

Maximizing over all $A \subseteq \mathcal{X}$ yields the result (see Exercise 3.7). □

Exercise 3.10: Pinsker's inequality for Bernoulli distributions

Show that

$$D_2(p\|q) \geq \frac{2}{\ln 2}(p-q)^2.$$

Hint: Show that the partial derivative of the function

$$f(p,q) = p\ln\frac{p}{q} + (1-p)\ln\frac{1-p}{1-q} - 2(p-q)^2$$

with respect to q is negative for $q \leq p$. The result follows since $f(p,p) = 0$.

We know that KL divergence *tensorizes* (see Theorem 3.6.3); namely, for product distributions P^n and Q^n, we have

$$D(P^n\|Q^n) = n \cdot D(P\|Q).$$

A similar property holds for total variation distance as well. More generally, we have

$$d_{\mathrm{TV}}\left(\prod_{i=1}^n P_i, \prod_{i=1}^n Q_i\right) \leq \sum_{i=1}^n d_{\mathrm{TV}}(P_i, Q_i).$$

Proof. We provide the proof for $n = 2$ (the general case follows by induction).

$$\begin{aligned}
2d_{\mathrm{TV}}(P_1P_2, Q_1Q_2) &= \sum_{x,y} |P_1(x)P_2(y) - Q_1(x)Q_2(y)| \\
&= \sum_{x,y} |P_1(x)P_2(y) - Q_1(x)Q_2(y) + P_1(x)Q_2(y) - P_1(x)Q_2(y)| \\
&\leq \sum_{x,y} |P_1(x)P_2(y) - P_1(x)Q_2(y)| + \sum_{x,y} |-Q_1(x)Q_2(y) + P_1(x)Q_2(y)| \\
&= \sum_x P_1(x) \sum_y |P_2(y) - Q_2(y)| + \sum_y Q_2(y) \sum_x |P_1(x) - Q_1(x)| \\
&= 2\left[d_{\mathrm{TV}}(P_2, Q_2) + d_{\mathrm{TV}}(P_1, Q_1)\right].
\end{aligned}$$

□

In particular, for product distributions P^n and Q^n, we have

$$d_{\mathrm{TV}}(P^n, Q^n) \leq n \cdot d_{\mathrm{TV}}(P, Q). \tag{3.5}$$

On the other hand, by Pinsker's inequality, we have

$$\begin{aligned}
d_{\mathrm{TV}}(P^n, Q^n) &\leq \sqrt{\frac{\ln 2}{2} \cdot D(P^n\|Q^n)} \\
&= \sqrt{\frac{\ln 2}{2} \cdot n \cdot D(P\|Q)}.
\end{aligned}$$

This is a tighter (smaller) upper bound than (3.5) when $d_{\mathrm{TV}}(P, Q)$ is of the same order as $\sqrt{D(P\|Q)}$; that is, when Pinsker's inequality is tight for P and Q.

3.5 Mutual information

Definition 3.16. For $(X, Y) \sim P_{XY}$ with marginals P_X and P_Y, the KL divergence between P_{XY} and P_XP_Y is referred to as the **mutual information** between X and Y; that is,

$$I(X \wedge Y) := D(P_{XY}\|P_XP_Y).$$

Observe that $I(X \wedge Y)$ is symmetric. It is easy to check the following:

$$\begin{aligned} I(X \wedge Y) &= H(X) - H(X|Y) \\ &= H(Y) - H(Y|X) \\ &= H(X) + H(Y) - H(X,Y). \end{aligned}$$

We know that $H(X)$ is the uncertainty in X and $H(X|Y)$ is the uncertainty remaining in X once Y is revealed. Thus, $I(X \wedge Y)$ is a measure of information revealed by Y about X (and vice versa).

Definition 3.17. The **conditional mutual information between** X **and** Y **given** Z is given by the expectation of the mutual information between X and Y with corresponding conditional distributions given $Z = z$. That is,

$$I(X \wedge Y|Z) := D\left(P_{XY|Z} \| P_{X|Z} P_{Y|Z} | P_Z\right).$$

Exercise 3.11: Properties of mutual information

Verify the following:

(a) $I(X \wedge Y) \geq 0$ with equality iff $X \perp\!\!\!\perp Y$

(b) $I(X \wedge X) = H(X)$

(c) $I(X \wedge Y) \leq \min\{H(X), H(Y)\}$

(d) $I(X \wedge Y_1, Y_2) = I(X \wedge Y_1) + I(X \wedge Y_2|Y_1)$.

To prove DPI for mutual information, we consider multiple stages of data processing which is modelled as a Markov chain. Formally,

Definition 3.18. $X - Y - Z$ forms a **Markov chain** if X and Z are independent given Y. The following conditions are equivalent to the definition:

i. $P_{XZ|Y} = P_{X|Y} P_{Z|Y}$

ii. $P_{Z|XY} = P_{Z|Y}$

iii. $P_{XYZ} = P_X P_{Y|X} P_{Z|Y}$.

Note: $P_{XZ|Y} = P_{X|Y} P_{Z|Y}$ means that for every $x \in \mathcal{X}$, $y \in \mathcal{Y}$, and $z \in \mathcal{Z}$,

$$P_{XZ|Y}(x, z|y) = P_{X|Y}(x|y) P_{Z|Y}(z|y).$$

Informally, an interpretation of the Markov chain $X - Y - Z$ is that Y is obtained by "processing" X and Z is obtained by (further) processing Y. Formally, however, a Markov chain provides the conditional independence structure and strictly speaking, not the direction of the flow of information. In this sense, it is not contradictory that if $X - Y - Z$ forms a Markov chain, so does $Z - Y - X$.
Notation: Sometimes, we simply say $X - Y - Z$ to mean "$X - Y - Z$ forms a Markov chain".

Example 3.3. $X - Y - Z \iff I(X \wedge Z|Y) = 0$.

Proof. We prove the forward direction first.

$$\begin{aligned} I(X \wedge Z|Y) &= H(X|Y) - H(X|Y,Z) \\ &= H(X|Y) - \mathbb{E}\left[-\log P_{X|Y,Z}\right] \\ &= H(X|Y) - \mathbb{E}\left[-\log P_{X|Y}\right] \\ &= H(X|Y) - H(X|Y) = 0. \end{aligned}$$

For the reverse direction, observe that since mutual information between X and Z given $Y = y$ is nonnegative, $I(X \wedge Z|Y) = 0$ implies that for every $y \in \mathcal{Y}$,

$$I(X \wedge Z|Y = y) = 0.$$

Then, by the property of mutual information (see Exercise 3.11.a), we conclude that the conditional distributions X given $Y = y$, and Z given $Y = y$ are independent for every $y \in \mathcal{Y}$. That is, $X - Y - Z$. □

The **DPI for mutual information** is as follows: If $X - Y - Z$, then,

$$I(X \wedge Z) \leq I(X \wedge Y)$$

with equality iff $X - Z - Y$.

Proof. By chain rule we have

$$I(Y, Z \wedge X) = I(X \wedge Y) + I(X \wedge Z|Y)$$

and

$$I(Y, Z \wedge X) = I(X \wedge Z) + I(X \wedge Y|Z)$$

Since $X - Y - Z$ implies $I(X \wedge Z|Y) = 0$ (see Example 3.3), the result follows. The condition for equality is also easy to check. □

3.5.1 Probability of error and Fano's inequality

Fano's inequality provides a lower bound on the probability of error while guessing a random variable from another (dependent) random variable. In particular, for the Markov chain $X - Y - \widehat{X}$, we seek to formalize the intuition that if the information that Y conveys about X is small, then the probability of error incurred while guessing X (to be $\widehat{X}$) using Y would be large. It can also be interpreted as a consequence of DPI for KL divergence but we use a simpler proof.

Theorem 3.10 (Fano's inequality)**.** *Let $X - Y - Z$ and $P_e = \mathbb{P}[X \neq \widehat{X}]$. Then,*

$$H(X|\widehat{X}) \leq P_e \log(|\mathcal{X}| - 1) + h(P_e)$$

where $h(\cdot)$ denotes the binary entropy function[2]*.*

Proof. Let

$$E := \{(x, \widehat{x}) : x \neq \widehat{x}\}.$$

Then,

$$\begin{aligned} H(X|\widehat{X}) &= H(X, \mathbf{1}_E|\widehat{X}) && \text{(since } \mathbf{1}_E \text{ is a function of } X, \widehat{X}) \\ &= H(\mathbf{1}_E|\widehat{X}) + H(X|\mathbf{1}_E, \widehat{X}) \\ &\leq H(\mathbf{1}_E) + H(X|\mathbf{1}_E, \widehat{X}) \\ &= h(P_e) + H(X|\mathbf{1}_E, \widehat{X}). \end{aligned}$$

We have

$$\begin{aligned} H(X|\mathbf{1}_E, \widehat{X}) &= P_{\mathbf{1}_E}(0) H(X|\mathbf{1}_E = 0, \widehat{X}) + P_{\mathbf{1}_E}(1) H(X|\mathbf{1}_E = 1, \widehat{X}) \\ &= P_e H(X|\mathbf{1}_E = 1, \widehat{X}) \\ &\leq P_e \log(|\mathcal{X}| - 1) \end{aligned}$$

which completes the proof. □

[2] For $0 \leq x \leq 1$, $h(x) = -x \log x - (1 - x) \log(1 - x)$.

Exercise 3.12: Lower bound on P_e

Show that

1. For $X \sim \mathrm{Unif}(\mathcal{X})$,
$$P_e \geq 1 - \frac{I(X \wedge Y) + 1}{\log |\mathcal{X}|}.$$

2. For $W - X - Y - \widehat{W}$ where $W \sim \mathrm{Unif}(\mathcal{X})$ with $p = \mathbb{P}[W \neq \widehat{W}]$,
$$p \geq 1 - \frac{I(X \wedge Y) + h(p)}{\log |\mathcal{X}|}.$$

Chapter 4

Sample complexity

For a given statistical inference problem, sample complexity refers to the smallest **number of samples necessary and sufficient** for solving the problem to within a desired accuracy, say ε. Often, sample complexity is characterized as a quantity *order-wise* dependent on ε and the parameters of the problem.

4.1 Binary hypothesis testing

We begin with a simple example.

How many coin tosses are needed to ***reliably*** *distinguish between a coin that always shows tails and a coin that shows tails with probability* $1-p$ *for* $p \approx 0$*?*

Formally, we have the following binary hypothesis testing problem:

$$\begin{aligned} &\mathcal{H}_0 : X^n \sim P^n \text{ where } P \equiv \mathrm{Ber}(0) && \text{; a coin that always shows tails} \\ &\mathcal{H}_1 : X^n \sim Q^n \text{ where } Q \equiv \mathrm{Ber}(p) && \text{; a coin that shows tails w.p. } 1-p. \end{aligned} \tag{4.1}$$

By "reliably", we mean that the test T that distinguishes between $\mathcal{H}_0$ and $\mathcal{H}_1$ should satisfy

$$P_e(T|P^n, Q^n) \leq \varepsilon. \tag{4.2}$$

That is, the Bayesian average probability of error with uniform priors (see Section 3.3.1) for the test must not exceed ε. The **sample complexity** of testing between $\mathcal{H}_0$ and $\mathcal{H}_1$ is the smallest number of samples n such that there exists a test T that uses n samples and satisfies (4.2). We denote it by $n_\varepsilon(P, Q)$.
Notation: We use subscripts P, Q and $\mathcal{H}_0, \mathcal{H}_1$ to indicate the distribution. For instance, $\mathrm{Var}_P(X)$ denotes the variance of $X \sim P$. Similarly, $\mathbb{P}_{\mathcal{H}_0}$ denotes the probability w.r.t. the distribution under $\mathcal{H}_0$.

First, we exhibit a test – the number of samples incurred by the test in order to satisfy (4.2) yields an upper bound on the sample complexity. Then, we provide a lower bound on the number of samples necessary for any test to satisfy (4.2).

(i) **(Upper bound)** Consider the test that takes as input the i.i.d. coin tosses $X_1, \ldots, X_n$ and has the following acceptance region:

$$\mathcal{A} = \{x^n = T^n = TTT \cdots T\}.$$

That is, the test declares $\mathcal{H}_0$ if X^n is the all-tail sequence and declares $\mathcal{H}_1$ for all other sequences. Then,

$$\begin{aligned} P_{\mathrm{FA}}(\mathcal{A}) &= 0, \\ P_{\mathrm{MD}}(\mathcal{A}) &= (1-p)^n. \end{aligned}$$

To ensure that (4.2) holds, it suffices to have

$$\frac{1}{2}(1-p)^n \leq \varepsilon.$$

Equivalently,

$$\begin{aligned} n &\geq \frac{\log 1/2\varepsilon}{\log 1/(1-p)} \\ &\geq \frac{1-p}{p} \log \frac{1}{2\varepsilon} \end{aligned}$$

where the last step follows by $\log(1+x) \leq x$ (see Exercise 1.13). Further, $p \approx 0 \iff 1-p \approx 1$. Thus, we have a test that uses $O(1/p)$ samples[1] and satisfies (4.2). Therefore, $n_\varepsilon(P,Q)$ is **at most** $O(1/p)$.

(ii) **(Lower bound)** For any test T satisfying (4.2), we must have

$$\begin{aligned} \varepsilon &\geq P_e(T|P^n, Q^n) \\ &\geq P_e(P^n, Q^n) \\ &= \frac{1}{2}\left[1 - d_{\mathrm{TV}}(P^n, Q^n)\right] \\ &\geq \frac{1}{2}(1-np)\,. \end{aligned}$$

the last step uses (see (3.5))

$$d_{\mathrm{TV}}(P^n, Q^n) \leq n \cdot d_{\mathrm{TV}}(P,Q).$$

Thus,

$$n \geq \frac{1-2\varepsilon}{p}.$$

That is, $n_\varepsilon(P,Q)$ is **at least** $\Omega(1/p)$.

Putting together the upper and the lower bounds, we conclude that:

the sample complexity of testing between the two coins is $\Theta(1/p)$.

Recall that for the example above, $d_{\mathrm{TV}}(P,Q) = p$. It is tempting to generalize and say that $n_\varepsilon(P,Q)$ is inversely proportional to $d_{\mathrm{TV}}(P,Q)$ for any distributions P and Q (on the binary alphabet if not more generally). However, this is not true. Consider the following example with P and Q with the same total variation distance.

How many coin tosses are needed to ***reliably*** *distinguish between a fair coin and a coin that favours heads slightly more than tails?*

Formally,

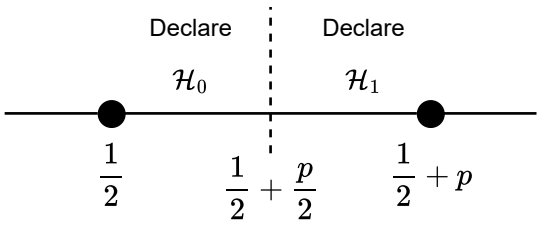
$$\begin{aligned} &\mathcal{H}_0 : X^n \sim P^n \text{ where } P \equiv \mathrm{Ber}\left(\frac{1}{2}\right) && \text{; a fair coin} \\ &\mathcal{H}_1 : X^n \sim Q^n \text{ where } Q \equiv \mathrm{Ber}\left(\frac{1}{2}+p\right) && \text{; a coin that favours H } \textit{slightly} \text{ more than T.} \end{aligned} \tag{4.3}$$

Intuitively, this problem seems harder than (4.1) and would demand many more samples to ensure the reliability (4.2). Indeed, we prove in the sequel that the sample complexity of (4.3) is $\Theta(1/p^2)$.

(i) **(Upper bound)** We use the following test: the test declares $\mathcal{H}_0$ if the fraction of heads falls to the left of the midpoint between the two Bernoulli parameters, namely, $\frac{1}{2}$ and $\frac{1}{2}+p$, and declares $\mathcal{H}_1$ otherwise (see figure). That is, the test has the following acceptance region:

$$\mathcal{A} = \left\{ x^n \in \{0,1\}^n : \frac{1}{n}\sum_{i=1}^n x_i \leq \frac{1}{2} + \frac{p}{2} \right\}.$$

[1] We ignore the dependence on ε for now.



Then,

$$\begin{aligned}
P_{\mathrm{FA}}(\mathcal{A}) &= P^n(\mathcal{A}^c) \\
&\leq P^n\left[\left|\frac{1}{n}\sum_{i=1}^n X_i - \frac{1}{2}\right| > \frac{p}{2}\right] \\
&\leq \frac{\mathrm{Var}_P(X_1)}{np^2/4} \\
&= \frac{1}{np^2}
\end{aligned}$$

where the penultimate step is by Chebyshev's inequality. The missed detection probability is upper bounded by

$$\begin{aligned}
P_{\mathrm{MD}}(\mathcal{A}) &= Q^n(\mathcal{A}) \\
&\leq Q^n\left[\left|\frac{1}{n}\sum_{i=1}^n X_i - \left(\frac{1}{2}+p\right)\right| > \frac{p}{2}\right] \\
&\leq \frac{\mathrm{Var}_Q(X_1)}{np^2/4} \\
&= \frac{\left(\frac{1}{2}+p\right)\left(\frac{1}{2}-p\right)}{np^2/4} \\
&\leq \frac{1}{np^2}.
\end{aligned}$$

Therefore, to ensure that (4.2) is satisfied, it suffices to have

$$n \geq \frac{1}{\varepsilon p^2}$$

and hence $n_\varepsilon(P,Q)$ is **at most** $O\left(1/p^2\right)$.

(ii) **(Lower bound)** Observe that since $d_{\mathrm{TV}}(P,Q)$ is the same as in (4.1), the lower bound is valid for this problem as well. However, the lower bound yields $\Omega(1/p)$ which (it so turns out) is weaker. We follow similar steps as before but instead of using the tensorization upper bound of the total variation distance, we use Pinsker's inequality (see Section 3.4.3). For any test T satisfying (4.2), we must have

$$\begin{aligned}
\varepsilon &\geq \frac{1}{2}\left[1 - d_{\mathrm{TV}}(P^n, Q^n)\right] \\
&\geq \frac{1}{2}\left(1 - \sqrt{\frac{\ln 2}{2} D(P^n \| Q^n)}\right) \\
&= \frac{1}{2}\left(1 - \sqrt{\frac{\ln 2}{2} \cdot n \cdot D(P\|Q)}\right)
\end{aligned}$$

which yields

$$n \geq \frac{2(1-2\varepsilon)^2}{\ln 2} \cdot \frac{1}{D(P\|Q)}.$$

We have (verify this)

$$D(P\|Q) = \frac{1}{2}\log\frac{1}{1-4p^2} \approx 2p^2$$

for small p and hence $n_\varepsilon(P,Q)$ is **at least** $\Omega(1/p^2)$.

Putting together the upper and the lower bounds, we conclude that:

the sample complexity of testing between the two coins is $\Theta(1/p^2)$.

More generally, we have the following theorem.

Theorem 4.1 (Sample complexity of binary hypothesis testing)**.** *Given i.i.d. samples $X^n \in \mathcal{X}^n$ from P^n under $\mathcal{H}_0$ and from Q^n under $\mathcal{H}_1$, for $0 < \varepsilon < 1/2$, where $P(x) = 0$ whenever $Q(x) = 0$ for all $x \in \mathcal{X}$, we have*

$$n_\varepsilon(P,Q) = \Theta\left(\frac{1}{\min\{D(P\|Q), D(Q\|P)\}}\right).$$

Proof. Consider the test with the following acceptance region.

$$\mathcal{A}_\delta = \left\{x^n \in \mathcal{X}^n : \frac{1}{n}\log\frac{P^n(x^n)}{Q^n(x^n)} \geq D(P\|Q) - \delta\right\}.$$

Observe that $\frac{P(x)}{Q(x)}$ is bounded and hence by Hoeffding's inequality, we have

$$P_{\text{FA}}(\mathcal{A}_\delta) = P^n(\mathcal{A}_\delta^c) \leq 2\exp\left\{-2n\delta^2\right\}. \tag{4.4}$$

Also,

$$\begin{aligned} P_{\text{MD}}(\mathcal{A}_\delta) &= Q^n(\mathcal{A}_\delta) \\ &= \sum_{x^n \in \mathcal{A}_\delta} Q^n(x^n) \\ &\leq \sum_{x^n \in \mathcal{A}_\delta} P^n(x^n) 2^{-n(D(P\|Q)+\delta)} \\ &\leq 2^{-n(D(P\|Q)-\delta)}. \end{aligned} \tag{4.5}$$

From (4.4)

$$n \geq \frac{1}{2\delta^2}\log\frac{2}{\varepsilon} \implies P_{\text{FA}}(\mathcal{A}_\delta) \leq \varepsilon.$$

From (4.5),

$$n \geq \frac{1}{D(P\|Q)-\delta}\log\frac{1}{\varepsilon} \implies P_{\text{MD}}(\mathcal{A}_\delta) \leq \varepsilon.$$

Letting $\delta \approx \sqrt{D(P\|Q)}$ allows us to conclude that $n_\varepsilon(P,Q)$ is at most $O\left(\frac{1}{D(P\|Q)}\right)$. The other term, namely, $D(Q\|P)$ is obtained using similar arguments. The lower bound is the same as in the case of (4.3). □

Exercise 4.1: Sample complexity of coin tosses

1. Verify that the general expression for $n_\varepsilon(P,Q)$ agrees with the expressions derived for (4.1) and (4.3).
2. Show that the sample complexity of distinguishing between a coin of bias p and a coin of bias $(1-\alpha)p$ where $\alpha \approx 0$ is $\Theta(1/\alpha p)$.

4.1.1 Refined sample complexity

In fact, we can derive the sample complexity for binary hypothesis testing in terms of n as well as ε using a small modification. Observe that the upper bound is actually

$$n_\varepsilon(P,Q) \le O\left(\frac{1}{D(P\|Q)}\log\frac{1}{\varepsilon}\right).$$

For the corresponding lower bound, we need the following result:

Lemma 4.1 (Bretagnolle-Huber inequality)**.** *For distributions P and Q,*

$$d_{\mathrm{TV}}^2(P,Q) \le 1 - \exp\{-D(P\|Q)\}.$$

To prove the inequality, we need the following identities which are easy to verify.

Exercise 4.2: Total variation distance identities

Show that

$$\begin{aligned} d_{\mathrm{TV}}(P,Q) &= 1 - \sum_{x\in\mathcal{X}} \min\{P(x),Q(x)\} \\ &= \sum_{x\in\mathcal{X}} \max\{P(x),Q(x)\} - 1. \end{aligned}$$

Proof. By the identities above, we have

$$\begin{aligned} 1 - d_{\mathrm{TV}}^2(P,Q) &= [1 + d_{\mathrm{TV}}(P,Q)][1 - d_{\mathrm{TV}}(P,Q)] \\ &= \left(\sum_{x\in\mathcal{X}} \max\{P(x),Q(x)\}\right)\left(\sum_{x\in\mathcal{X}} \min\{P(x),Q(x)\}\right) \\ &\ge \left(\sum_{x\in\mathcal{X}} \sqrt{\max\{P(x),Q(x)\}\min\{P(x),Q(x)\}}\right)^2 \\ &= \left(\sum_{x\in\mathcal{X}} \sqrt{P(x)Q(x)}\right)^2 \end{aligned}$$

where the inequality is by Cauchy-Schwarz. By Jensen's inequality, we have

$$\begin{aligned} \left(\sum_{x\in\mathcal{X}} \sqrt{P(x)Q(x)}\right)^2 &= \exp\left\{2\ln\sum_{x\in\mathcal{X}} \sqrt{P(x)Q(x)}\right\} \\ &= \exp\left\{2\ln\sum_{x\in\mathcal{X}} P(x)\sqrt{\frac{Q(x)}{P(x)}}\right\} \\ &= \exp\left\{2\ln\mathbb{E}_P\sqrt{\frac{Q(X)}{P(X)}}\right\} \\ &\ge \exp\left\{-2\mathbb{E}_P\ln\sqrt{\frac{P(X)}{Q(X)}}\right\} \\ &= \exp\{-D(P\|Q)\} \end{aligned}$$

which completes the proof. □

Next, we use the Bretagnolle-Huber inequality to provide a refined lower bound for the sample complexity of testing between P and Q. Consider any test T satisfying (4.2). Then,

$$\begin{aligned}\varepsilon &\geq \frac{1}{2}\left[1 - d_{\mathrm{TV}}(P^n, Q^n)\right] \\ &\geq \frac{1}{2}\left[1 - \sqrt{1 - e^{-nD(P\|Q)}}\right]\end{aligned}$$

by Lemma 4.1. Thus,

$$\begin{aligned}n &\geq \frac{1}{D(P\|Q)} \log \frac{1}{1-(1-2\varepsilon)^2} \\ &\geq \frac{1}{D(P\|Q)} \log \frac{1}{4\varepsilon}\end{aligned}$$

which allows us to conclude that

$$n_\varepsilon(P, Q) = \Theta\left(\frac{1}{\min\{D(P\|Q), D(Q\|P)\}} \log \frac{1}{\varepsilon}\right).$$

Exercise 4.3: Weaker "Bretagnolle-Huber" inequality

Using the Bretagnolle-Huber inequality, show that

$$d_{\mathrm{TV}}(P, Q) \leq 1 - \frac{1}{2} e^{-D(P\|Q)}.$$

Use this inequality to derive the same bound on n as above.

4.2 Estimation

Another canonical problem in statistics, apart from hypothesis testing, is that of **estimation**. Suppose we observe $Y \sim P_\theta$ where P_θ is a probability distribution parameterized by $\theta \in \Lambda$. For example, $Y \sim \mathrm{Ber}(\theta)$; $\theta \in [0, 1]$. Having observed Y, we seek to give a "best guess" for θ; call it the *estimator* $\widehat{\theta}(Y)$. Note that $\widehat{\theta} : \mathcal{Y} \to \Lambda$. The value of $\widehat{\theta}(y)$ for a particular $Y = y$ is called the *estimate* of Θ.

The performance of the estimator is measured using a cost function

$$C : \Lambda \times \Lambda \to \mathbb{R}^+$$

where $C(\widehat{\theta}(y), \theta)$ is the cost of $\widehat{\theta}(y)$ when y is observed and θ is the true parameter. A few example cost functions are shown in Figure 4.1.

Suppose the true parameter is $\theta \in \Lambda$ and $Y \sim P_\theta$. We seek to find a $\widehat{\theta}(Y)$ such that

$$R_\theta(\widehat{\theta}(Y)) := \mathbb{E}_{P_\theta}[C(\widehat{\theta}(Y), \theta)] = \int_{\mathcal{Y}} P_\theta(y) C(\widehat{\theta}(y), \theta) dy$$

is minimized. The quantity $R_\theta(\widehat{\theta}(Y))$ is called the conditional risk.

4.2.1 Bayesian framework

Akin to Bayesian hypothesis testing where we assumed prior distributions on the hypotheses, here we assume that the parameter θ is random and is distributed on Λ according to a PDF $w(\theta)$. Then, the **Bayes' risk** is given by

$$\begin{aligned}r(\widehat{\theta}) &:= \mathbb{E}_{\Theta \sim w(\theta)}\left[R_\Theta(\widehat{\theta}(Y))\right] \\ &= \mathbb{E}_{w(\theta)}\left[\mathbb{E}_{P_\Theta}\left[C(\widehat{\theta}(Y), \Theta)\right]\right] \\ &= \int_\Lambda w(\theta) \int_{\mathcal{Y}} P_\theta(y) C(\widehat{\theta}(y), \theta) dy \cdot d\theta.\end{aligned}$$

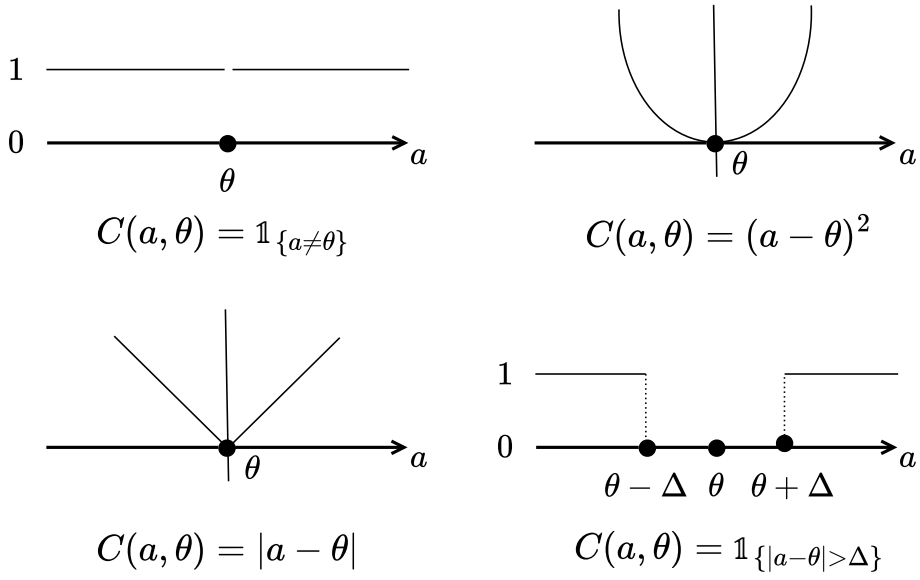


Figure 4.1: Examples of cost functions

Our goal is to find $\widehat{\theta}(\cdot)$ that minimizes the Bayes' risk $r(\widehat{\theta})$. Observe that

$$\begin{aligned} r(\widehat{\theta}) &= \mathbb{E}\left[C(\widehat{\theta}(Y), \Theta)\right] \\ &= \mathbb{E}_Y\left[\mathbb{E}_{\Theta|Y}\left[C(\widehat{\theta}(y), \Theta)|Y = y\right]\right] \end{aligned}$$

where the conditional distribution has PDF

$$w(\theta|y) = \frac{w(\theta)P_\theta(y)}{P(y)} \tag{4.6}$$

with

$$P(y) = \int_\Lambda w(\theta)P_\theta(y)d\theta.$$

To minimize $r(\widehat{\theta})$, we solve the smaller problem for $Y = y$ first. That is, for $Y = y$, we seek $\widehat{\theta}(y)$ where

$$\begin{aligned} \widehat{\theta}(y) &= \arg\min_{a\in\Lambda} \mathbb{E}_{\Theta|Y}\left[C(a, \Theta)|Y = y\right] \\ &= \arg\min_{a\in\Lambda} \int_\Lambda C(a, \theta)w(\theta|y)d\theta. \end{aligned}$$

Next, we analyze the form of $\widehat{\theta}(y)$ for different cost functions.

Theorem 4.2. *Let $\Lambda = \mathbb{R}$ and let $\mathbb{E}\Theta^2 < \infty$. For the squared error cost function*

$$C(a, \theta) = (a - \theta)^2,$$

we have

$$\widehat{\theta}(y) = \mathbb{E}[\Theta|Y = y].$$

The estimator $\widehat{\theta}(y)$ is called the **Minimum Mean Squared Error (MMSE)** estimator and it is the **conditional mean** of Θ given $Y = y$. Observe that the minimum mean squared error is given by

$$\mathbb{E}\left[\left(\widehat{\Theta}(Y) - \Theta\right)^2\right] = \mathbb{E}_Y\left[\mathbb{E}_{\Theta|Y}\left[(\mathbb{E}[\Theta|Y] - \Theta)^2\right]\right] = \mathbb{E}\left[\mathrm{Var}\left(\Theta|Y\right)\right].$$

Proof. Given $Y = y$, let $a = \widehat{\theta}(y)$ and let $\mu = \mathbb{E}[\Theta|Y = y]$. Our goal is to choose a to minimize

$$\begin{aligned}\mathbb{E}[(a-\Theta)^2|Y=y] &= \mathbb{E}[(a-\mu+\mu-\Theta)^2|Y=y] \\ &= (a-\mu)^2 + \mathbb{E}[(\mu-\Theta)^2|Y=y] + 2(a-\mu)\mathbb{E}[\mu-\Theta|Y=y] \\ &= (a-\mu)^2 + \mathbb{E}[(\mu-\Theta)^2|Y=y] \\ &\geq \mathbb{E}[(\mu-\Theta)^2|Y=y]\end{aligned}$$

where equality is achieved when $a = \mu$. □

Theorem 4.3. *Let $\Lambda = \mathbb{R}$ and let $\mathbb{E}\Theta^2 < \infty$. For the absolute error cost function*

$$C(a,\theta) = |a-\theta|,$$

$\widehat{\theta}(y)$ is any point on $\mathbb{R}$ such that

$$a < \widehat{\theta}(y) \Longrightarrow \mathbb{P}\left[\Theta < a|Y=y\right] \leq \mathbb{P}\left[\Theta > a|Y=y\right]$$

and

$$a > \widehat{\theta}(y) \Longrightarrow \mathbb{P}\left[\Theta < a|Y=y\right] \geq \mathbb{P}\left[\Theta > a|Y=y\right].$$

That is, $\widehat{\theta}(y)$ is the **Minimum Mean Absolute Error (MMAE)** estimator and it is the **conditional median** of Θ given $Y = y$.

Exercise 4.4: Continuous analogue of Theorem 1.4

Show that if X is a non-negative random variable, then

$$\mathbb{E}X = \int_0^\infty \mathbb{P}[X > x]dx.$$

Hint: Write $\mathbb{E}X = \int_0^\infty x f_X(x)dx$ with $x = \int_0^x 1dy$.

Proof. Using the result in Exercise 4.4, we can write

$$\begin{aligned}\mathbb{E}\left[|\Theta-a|\middle|Y=y\right] &= \int_0^\infty \mathbb{P}\left[|\Theta-a|>t\middle|Y=y\right]dt \\ &= \int_0^\infty \mathbb{P}\left[\Theta>a+t\middle|Y=y\right]dt + \int_0^\infty \mathbb{P}\left[\Theta<a-t\middle|Y=y\right]dt \\ &= \int_a^\infty \mathbb{P}\left[\Theta>u\middle|Y=y\right]du + \int_{-\infty}^a \mathbb{P}\left[\Theta<u\middle|Y=y\right]du.\end{aligned}$$

Therefore,

$$\frac{d}{da}\mathbb{E}\left[|\Theta-a|\middle|Y=y\right] = -\mathbb{P}\left[\Theta>a\middle|Y=y\right] + \mathbb{P}\left[\Theta<a\middle|Y=y\right] \tag{4.7}$$

$$= -1 + \mathbb{P}\left[\Theta\leq a\middle|Y=y\right] + \mathbb{P}\left[\Theta<a\middle|Y=y\right]. \tag{4.8}$$

Using (4.8), we can observe that the derivative is an increasing function of a. The minimum is attained at any point where to the left, the slope is negative and to the right, it is positive. From (4.7), this is satisfied for $\widehat{\theta}(y)$ specified in the Theorem statement. □

Finally, we consider the limit of another kind of cost function.

Theorem 4.4. *Let* $\Lambda = \mathbb{R}$. *Let*

$$C(a, \theta) = \mathbf{1}_{\{|a-\theta|>\Delta\}}.$$

Then,

$$\lim_{\Delta \to 0} \frac{\mathbb{E}[C(a, \Theta)|Y = y] - 1}{\Delta}$$

is minimized at

$$\widehat{\theta}(y) = \arg\max_a w(a|y).$$

This estimator is called the **maximum a posteriori probability (MAP)** estimator and it is the **conditional mode** of the posterior distribution (of Θ given $Y = y$).

Proof. We have

$$\begin{aligned}
\mathbb{E}[C(\widehat{\theta}(y), \Theta)|Y = y] &= \mathbb{P}[|\widehat{\theta}(y) - \Theta| > \Delta|Y = y] \\
&= 1 - \mathbb{P}[|\widehat{\theta}(y) - \Theta| \leq \Delta|Y = y] \\
&= 1 - \mathbb{P}\left[\Theta \in \left(\widehat{\theta}(y) - \Delta, \widehat{\theta}(y) + \Delta\right) \middle| Y = y\right].
\end{aligned}$$

Therefore,

$$\begin{aligned}
\lim_{\Delta \to 0} \frac{\mathbb{E}[C(\widehat{\theta}(y), \Theta)|Y = y] - 1}{\Delta} &= \lim_{\Delta \to 0} \frac{-1}{\Delta} \mathbb{P}\left[\Theta \in \left(\widehat{\theta}(y) - \Delta, \widehat{\theta}(y) + \Delta\right) \middle| Y = y\right] \\
&= \lim_{\Delta \to 0} \frac{-1}{\Delta} \cdot 2\Delta \cdot w(\widehat{\theta}(y)|y) \\
&= -2w(\widehat{\theta}(y)|y).
\end{aligned}$$

The choice of $\widehat{\theta}(y)$ that minimizes the above expression is given by

$$\widehat{\theta}(y) = \arg\max_a w(a|y).$$

□

Exercise 4.5: MMSE

1. Let $X \sim \text{Ber}(\theta)$ where θ is uniformly distributed over $[0, 1]$. Derive the MMSE estimate of Θ given X and calculate the mean squared error.

2. Let $\Theta \sim \exp(\alpha)$. That is,

$$w(\theta) = \alpha e^{-\alpha\theta}, \theta > 0,$$

and for a fixed θ, $Y \sim \exp(\theta)$. That is,

$$P_\theta(y) = \theta e^{-\theta y}, y > 0.$$

Calculate the MMSE estimate of Θ given $Y = y$.

Observe that by Bayes' rule (see (4.6)), the MAP estimator is

$$\widehat{\theta}(y) = \arg\max_a w(a|y) = \arg\max_\theta w(\theta) P_\theta(y).$$

When we do not know the prior distribution of the parameter θ ($w(\theta)$ is unknown) we may assume that θ is uniformly distributed on Λ and the MAP estimator becomes

$$\widehat{\theta}(y) = \arg\max_\theta P_\theta(y)$$

which is known as the **Maximum Likelihood (ML)** estimator.

Exercise 4.6: MLE

1. Given $X_1, \ldots, X_n$ i.i.d. $\mathrm{Ber}(p)$, compute the ML estimate of p.
2. Given $X_1, \ldots, X_n$ be i.i.d. $\mathrm{Geo}(p)$, compute the ML estimate of p.
3. Given $X_1, \ldots, X_n$ be i.i.d. $\mathrm{Unif}[m]$, compute the ML estimate of m.

4.2.2 Minmax framework

Given i.i.d. samples $X_1, \ldots, X_n$ of coin tosses, we seek to characterize the minmax risk for estimation of the bias of the coin with squared error cost function. The risk of an estimator $\widehat{p}(X^n)$ is given by

$$R_n(\widehat{p}) = \max_{p \in [0,1]} \mathbb{E}_p \left[|\widehat{p}(X^n) - p|^2 \right]$$

and the minmax risk is given by

$$R_n = \min_{\widehat{p}} R_n(\widehat{p}).$$

Our goal is to characterize R_n as a function of the number of samples n.

Theorem 4.5. *Given $X_1, \ldots, X_n$ i.i.d. $\mathrm{Ber}(p)$, the minmax risk of estimating $p \in [0,1]$ is*

$$R_n = \Theta\left(\frac{1}{n}\right).$$

Proof. (i) **(Upper bound)** We use the ML estimator (see Exercise 4.5.3):

$$\widehat{p}(X^n) = \frac{1}{n} \sum_{i=1}^{n} X_i.$$

Then,

$$\mathbb{E}_p \left[|\widehat{p}(X^n) - p|^2 \right] = \mathbb{E}_p \left[\left| \frac{1}{n} \sum_{i=1}^{n} X_i - p \right|^2 \right] = \frac{p(1-p)}{n}$$

whereby

$$R_n(\widehat{p}) = \max_{p \in [0,1]} \frac{p(1-p)}{n} \leq \frac{1}{4n}.$$

Therefore, R_n is **at most** $O(1/n)$.

(ii) **(Lower bound)** We derive a lower bound on R_n via a reduction to a binary hypothesis testing problem. This is known as **Le Cam's method**. For any estimator $\widehat{p}$, we have

$$R_n(\widehat{p}) = \max_{p \in [0,1]} \mathbb{E}_p \left[|\widehat{p}(X^n) - p|^2 \right] \geq \max_{j \in \{0,1\}} \mathbb{E}_{p_j} \left[|\widehat{p}(X^n) - p_j|^2 \right] \tag{4.9}$$

where we fix p_0 and p_1 such that

$$|p_0 - p_1| = \delta. \tag{4.10}$$

Consider the following hypothesis testing problem with observations X^n i.i.d. $\mathrm{Ber}(p_j)$ under $\mathcal{H}_j, j = 0, 1$. We use test T that outputs j if $|\widehat{p}(X^n) - p_j| < \frac{\delta}{2}$ for $j = 0, 1$. Then,

$$\begin{aligned} \mathbb{P}_{p_j}[T(X^n) \neq j] &\leq \mathbb{P}_{p_j} \left[|\widehat{p}(X^n) - p_j| \geq \frac{\delta}{2} \right] \\ &\leq \mathbb{E}_{p_j} \left[|\widehat{p}(X^n) - p_j|^2 \right] \cdot \frac{4}{\delta^2}. \end{aligned} \tag{4.11}$$

by Markov's inequality. From (4.9) and (4.11), we have

$$\begin{aligned} R_n(\widehat{p}) &\geq \frac{\delta^2}{4} \max_{j\in\{0,1\}} \mathbb{P}_{p_j}[T(X^n) \neq j] \\ &\geq \frac{\delta^2}{4} \cdot \frac{1}{2} \left[\mathbb{P}_{p_0}[T(X^n) \neq 0] + \mathbb{P}_{p_1}[T(X^n) \neq 1]\right] \\ &= \frac{\delta^2}{4} \cdot P_e(T|p_0^n, p_1^n) \\ &\geq \frac{\delta^2}{4} \cdot P_e(p_0^n, p_1^n). \end{aligned}$$

where the second inequality is by "max $\geq$ average". Then, using Pinsker's inequality like before (see lower bound for sample complexity of (4.3)), we get

$$R_n(\widehat{p}) \geq \frac{\delta^2}{4} \cdot \left[1 - \sqrt{\frac{\ln 2}{2} \cdot nD_2(p_0\|p_1)}\right]$$

where $D_2(p_0\|p_1)$ denotes the KL divergence between $\mathrm{Ber}(p_0)$ and $\mathrm{Ber}(\mathrm{p}_1)$. We choose $p_0 = 1/2$ and $p_1 = 1/2 + \delta$ so that (4.10) is satisfied. Recall from the calculation of lower bound for sample complexity of (4.3) that for this choice,

$$D_2(p_0\|p_1) = \frac{1}{2} \log \frac{1}{1 - 4\delta^2} \approx 2\delta^2.$$

Thus,

$$R_n(\widehat{p}) \geq \frac{\delta^2}{4} \cdot \left[1 - \sqrt{n\delta^2 \ln 2}\right].$$

We choose δ to satisfy $n\delta^2 \ln 2 = 1/4$ so that

$$R_n(\widehat{p}) \geq \frac{1}{32n \ln 2}.$$

Thus, R_n is **at least** $\Omega(1/n)$.

□

4.2.3 PAC framework

Here, we consider the Probably Approximately Correct (PAC) framework for the problem of estimating the bias of a coin given i.i.d. tosses with

$$\epsilon(n, \delta) := \min_{\widehat{p}} \max_{p\in[0,1]} \mathbb{P}_p\left[|\widehat{p}(X^n) - p| > \delta\right]$$

and the PAC sample complexity defined by

$$n_\varepsilon(\delta) := \min\{n : \epsilon(n, \delta) \leq \varepsilon\}.$$

Theorem 4.6. *Given $X_1, \ldots, X_n$ i.i.d. $\mathrm{Ber}(p)$, the PAC sample complexity of estimating $p \in [0, 1]$ is*

$$n_\varepsilon(\delta) = \Theta\left(\frac{1}{\delta^2} \log \frac{1}{\varepsilon}\right).$$

Proof. (i) **(Upper bound)** For the estimator

$$\widehat{p}(X^n) = \frac{1}{n} \sum_{i=1}^{n} X_i,$$

by Hoeffding's inequality, we have

$$\mathbb{P}_p\left[|\widehat{p}(X^n) - p| > \delta\right] \leq 2e^{-2n\delta^2}.$$

Thus, to ensure that

$$\max_{p\in[0,1]} \mathbb{P}_p\left[|\widehat{p}(X^n) - p| > \delta\right] \leq \varepsilon$$

it suffices to have

$$n \geq \frac{1}{2\delta^2}\log\frac{2}{\varepsilon}.$$

Hence, $n_\varepsilon(\delta)$ is **at most** $O\left(\frac{1}{\delta^2}\log\frac{1}{\varepsilon}\right)$.

(ii) **(Lower bound)** Given an estimator $\widehat{p}$ such that

$$\mathbb{P}_p\left[|\widehat{p}(X^n) - p| > \delta\right] \leq \varepsilon$$

for all $p \in [0,1]$, we have for $p_0 = 1/2$ and $p_1 = 1/2 + 2\delta$,

$$\frac{1}{2}\Big(\mathbb{P}_{p_0}\left[|\widehat{p}(X^n) - p_0| > \delta\right] + \mathbb{P}_{p_1}\left[|\widehat{p}(X^n) - p_1| > \delta\right]\Big) \leq \varepsilon.$$

That is, for the hypothesis testing problem with X^n i.i.d. p_j under $\mathcal{H}_j, j = 0,1$, for the test T that declares $\mathcal{H}_j$ if $|\widehat{p}(X^n) - p_j| < \delta$, we have

$$\varepsilon \geq P_e(T|p_0^n, p_1^n).$$

Hence,

$$\varepsilon \geq P_e(p_0^n, p_1^n)$$

and using the Bretagnolle-Huber inequality (Lemma 4.1), we obtain $n_\varepsilon(\delta)$ is **at least** $\Omega\left(\frac{1}{\delta^2}\log\frac{1}{\varepsilon}\right)$.

□

Exercise 4.7: Lower bound

Verify the lower bound above using the Bretagnolle-Huber inequality.

4.3 Learning discrete distributions

We study the sample complexity of learning a distribution from the samples in the PAC framework. We observe samples $X_1, \ldots, X_n$ i.i.d. from some unknown distribution P and the goal is to output $\widehat{P}$ such that $d_{\mathrm{TV}}(P, \widehat{P}) \leq \delta$ w.p. at least $1 - \varepsilon$. That is,

$$\mathbb{P}_{X^n\sim P^n}\left[d_{\mathrm{TV}}(\widehat{P}(X^n), P) \geq \delta\right] \leq \varepsilon. \tag{4.12}$$

Let the size of the alphabet from which the sample X_i are drawn be k; that is, $|\mathcal{X}| = k$. The sample complexity $n(k, \delta, \varepsilon)$, the smallest number of samples for which there exists an estimator that satisfies (4.12), is characterized in the following theorem.

Theorem 4.7.

$$n(k, \delta, \varepsilon) = \Theta\left(\frac{k + \log(1/\varepsilon)}{\delta^2}\right).$$

Proof. (i) **(Upper bound)** Consider the empirical distribution

$$\widehat{P}(j) = \frac{1}{n}\sum_{i=1}^{n} \mathbf{1}_{\{X_i = j\}}; \qquad j \in [k].$$

Note that $\widehat{P}$ is a random variable, and so is $d_{\mathrm{TV}}(\widehat{P}, P)$. We bound the expected value of the latter.

$$\begin{aligned}\mathbb{E}\left[d_{\mathrm{TV}}(\widehat{P}, P)\right] &= \frac{1}{2}\sum_{j=1}^{k}\mathbb{E}\left[\left|\widehat{P}(j) - P(j)\right|\right] \\ &\leq \frac{1}{2}\sum_{j=1}^{k}\sqrt{\mathbb{E}\left[\left(\widehat{P}(j) - P(j)\right)^2\right]} \end{aligned} \tag{4.13}$$

by Cauchy-Schwarz inequality. For every $j \in [k]$, the quantity $n\widehat{P}(j) \sim \mathrm{Bin}(n, P(j))$ whereby

$$\begin{aligned}\mathbb{E}\left[\left(\widehat{P}(j) - P(j)\right)^2\right] &= \frac{1}{n^2}\mathrm{Var}\left(n\widehat{P}(j)\right) \\ &= \frac{1}{n^2}\cdot nP(j)(1 - P(j)) \\ &\leq \frac{P(j)}{n}.\end{aligned}$$

Substituting this into (4.19) and applying Jensen's inequality yields

$$\begin{aligned}\mathbb{E}\left[d_{\mathrm{TV}}(\widehat{P}, P)\right] &\leq \frac{1}{2\sqrt{n}}\sum_{j=1}^{k}\sqrt{P(j)} \\ &= \frac{1}{2\sqrt{n}}\cdot k\sum_{j=1}^{k}\frac{1}{k}\sqrt{P(j)} \\ &\leq \frac{1}{2\sqrt{n}}\cdot k\sqrt{\sum_{j=1}^{k}\frac{P(j)}{k}} \\ &= \frac{1}{2}\sqrt{\frac{k}{n}}.\end{aligned} \tag{4.14}$$

To ensure that $\mathbb{E}\left[d_{\mathrm{TV}}(\widehat{P}, P)\right] \leq \frac{\delta}{2}$, it suffices to have

$$n \geq \frac{k}{\delta^2}. \tag{4.15}$$

The final step is to convert the bound on expectation into a large probability bound. Observe that the function

$$f(X_1, \ldots, X_n) := d_{\mathrm{TV}}(\widehat{P}(X^n), P)$$

satisfies the bounded differences property (see Definition 2.2) with $c_i = 2/n$ for all $i \in [n]$. Therefore, by McDiarmid's inequality (Theorem 2.5), we have

$$\mathbb{P}\left[f(X^n) - \mathbb{E}f(X^n) \geq \frac{\delta}{2}\right] \leq \exp\left\{-\frac{n\delta^2}{32}\right\} \tag{4.16}$$

Hence, from (4.14), for n satisfying (4.15), we have

$$\begin{aligned}\mathbb{P}\left[d_{\mathrm{TV}}(\widehat{P}(X^n), P) \geq \delta\right] &\leq \mathbb{P}\left[d_{\mathrm{TV}}(\widehat{P}(X^n), P) - \frac{\delta}{2} \geq \frac{\delta}{2}\right] \\ &\leq \mathbb{P}\left[d_{\mathrm{TV}}(\widehat{P}(X^n), P) - \mathbb{E}\left[d_{\mathrm{TV}}(\widehat{P}(X^n), P)\right] \geq \frac{\delta}{2}\right].\end{aligned} \tag{4.17}$$

Thus, from (4.17) and (4.16), to ensure that (4.12) is satisfied, it suffices to have

$$n \geq \frac{32}{\delta^2} \log \frac{1}{\varepsilon}. \tag{4.18}$$

From (4.15) and (4.18), we have

$$n(k, \varepsilon, \delta) \leq O\left(\max\left\{\frac{k}{\delta^2}, \frac{1}{\delta^2}\log\frac{1}{\varepsilon}\right\}\right) \leq O\left(\frac{k + \log\frac{1}{\varepsilon}}{\delta^2}\right)$$

Exercise 4.8: Alternative argument

Can you derive a bound similar to (4.18) using Hoeffding's inequality (Theorem 2.3) instead of McDiarmid's?

(ii) **(Lower bound)** We provide the lower bound for the dependence on k only and omit the one for the dependence on ε. The strategy is to pick a family of distributions and show that learning a uniformly randomly chosen distribution from the family while guaranteeing that (4.12) holds with expected total variation distance (since the true distribution is random) requires $\Omega(k/\delta^2)$ samples. This allows us to conclude that there exist distributions for which learning them requires so many samples.

Assume WLOG that k is even. For the pair $\{j, j+1\}$ where $j \in \{1, 3, \ldots, k-1\}$, toss a fair coin $Z_j \in \{+1, -1\}$ and set

$$p_j = \frac{1 - \delta Z_j}{k},$$
$$p_{j+1} = \frac{1 + \delta Z_j}{k}.$$

We refer to the family of distributions generated in this manner as $\mathcal{S}$. Note that $|\mathcal{S}| = 2^{k/2}$. We refer to the pair $\{j, j+1\}$ as the *bucket* j for $j \in \{1, 3, \ldots, k-1\}$. Let p_z be the true distribution and suppose the learning algorithm outputs $\widehat{p}$. Then, by guarantee (4.12),

$$d_{\mathrm{TV}}(\widehat{p}, p_z) \leq \delta \qquad \text{w.h.p.}$$

We can always find a distribution $p_{z'} \in \mathcal{S}$ that is δ-close to $\widehat{p}$ (in total variation distance). For instance, p_z is one such distribution. Then, by triangle inequality,

$$d_{\mathrm{TV}}(p_{z'}, p_z) \leq 2\delta \qquad \text{w.h.p.}$$

Therefore, estimating $Z \in \{+1, -1\}^{k/2}$ results in learning p_Z and learning p_Z provides us with an estimate for Z. Hence, we assume that the learning algorithm takes X^n as input and outputs $W \in \{+1, -1\}^{k/2}$, an estimate for Z according to which p_Z is designed. Let $Z \sim \mathrm{Unif}\{+1, -1\}^{k/2}$ and let $X \in [k]^n$ be $(X_1, \ldots, X_n)$. Then,

$$d_{\mathrm{TV}}(p_Z, p_W) = \frac{4\delta}{k} \sum_{j=1}^{k/2} \mathbf{1}\{W_j \neq Z_j\}.$$

Let B_i denote the bucket in which the sample X_i falls. Then,

$$\begin{aligned}
\mathbb{E}\left[d_{\mathrm{TV}}(p_W, p_Z)\right] &= \frac{4\delta}{k} \sum_{j=1}^{k/2} \mathbb{E}\left[\mathbf{1}\{W_j(X) \neq Z_j\}\right] \\
&= \frac{4\delta}{k} \sum_{j=1}^{k/2} \mathbb{E}\left[\mathbb{E}\left[\mathbf{1}\{W_j(X) \neq Z_j\} \middle| B_1, \ldots, B_n\right]\right]. \qquad (4.19)
\end{aligned}$$

$W_j(X)$ depends only on the N_j samples that fall in bucket j; i.e., the samples for which $B_i = j$ and

$$N_j = \sum_{i=1}^{n} \mathbf{1}_{\{B_i=j\}}.$$

The conditional distribution of X_i given B_i is supported on $\{B_i, B_i+1\}$ with PMF $\left(\frac{1-\delta Z_{B_i}}{2}, \frac{1+\delta Z_{B_i}}{2}\right)$. Therefore, conditioned on B^n, W_j is the output of a test T_j that uses $N_j = n_j$ samples to distinguish between the distributions

$$P \equiv \left(\frac{1-\delta}{2}, \frac{1+\delta}{2}\right) \text{ and } Q \equiv \left(\frac{1+\delta}{2}, \frac{1-\delta}{2}\right)$$

with

$$D(P\|Q) \approx 2\delta^2. \qquad \text{(verify this)}$$

Thus,

$$\mathbb{E}\left[\mathbf{1}\{W_j(X) \neq Z_j\} \middle| B_1, \ldots, B_n\right] = P_e(T_j | P^{n_j}, Q^{n_j})$$

and hence using arguments like before (Pinsker's inequality, tensorization etc.) we obtain

$$\mathbb{E}\left[\mathbf{1}\{W_j(X) \neq Z_j\} \middle| B_1, \ldots, B_n\right] \geq \frac{1}{2}\left[1 - \sqrt{n_j \delta^2 \ln 2}\right]. \tag{4.20}$$

Since B_i is uniform over the odd integers between 1 and $k-1$, we have

$$\mathbb{E}N_j = \sum_{i=1}^{n} \mathbb{P}[B_i = j] = \frac{2n}{k}$$

and hence by Jensen's inequality,

$$\mathbb{E}\sqrt{N_j} \leq \sqrt{\mathbb{E}N_j} = \sqrt{\frac{2n}{k}}.$$

Substituting the above in (4.20) yields

$$\mathbb{E}\left[\mathbb{E}\left[\mathbf{1}\{W_j(X) \neq Z_j\} \middle| B_1, \ldots, B_n\right]\right] \geq \frac{1}{2}\left[1 - \delta\sqrt{\frac{2n \ln 2}{k}}\right] \tag{4.21}$$

whereby from (4.19), we have

$$\mathbb{E}\left[d_{\mathrm{TV}}(p_W, p_Z)\right] \geq \delta\left[1 - \delta\sqrt{\frac{2n \ln 2}{k}}\right]. \tag{4.22}$$

By the assumption (4.12) on the estimator (with $\mathbb{E}_Z\left[d_{\mathrm{TV}}(p_W, p_Z)\right]$ in place of $d_{\mathrm{TV}}(\widehat{P}, P)$), w.p. at least $1-\varepsilon$ (with respect to random X^n), we must have

$$\mathbb{E}_Z\left[d_{\mathrm{TV}}(p_W, p_Z)\right] \leq \delta.$$

Let Y denote $\mathbb{E}_Z\left[d_{\mathrm{TV}}(p_W, p_Z)\right]$. Taking expectation of $d_{\mathrm{TV}}(p_W(X^n), p_Z)$ w.r.t. (Z, X^n), by Exercise 4.4, we have

$$\begin{aligned}
\mathbb{E}\left[d_{\mathrm{TV}}(p_W, p_Z)\right] &= \mathbb{E}Y \\
&= \int_0^{\delta} \mathbb{P}[Y \geq y]dy + \int_{\delta}^{1} \mathbb{P}[Y \geq y]dy \\
&\leq y\Big|_0^{\delta} + \varepsilon y\Big|_{\delta}^{1} \\
&= \delta + \varepsilon - \varepsilon\delta \\
&\leq c_\varepsilon \delta
\end{aligned} \tag{4.23}$$

for some $0 < c_\varepsilon < 1$ for ε small enough. From (4.22) and (4.23), we need

$$\delta \left[1 - \delta \sqrt{\frac{2n \ln 2}{k}} \right] \leq c_\varepsilon \delta,$$

equivalently,

$$n \geq \frac{ck}{\delta^2}.$$

□

The branch of **property testing** deals with the following:

Given i.i.d. samples from an unknown distribution, can we efficiently decide whether the distribution follows some "property", or is it statistically far from doing so?

Some examples are whether the distribution is *unimodal*, whether the distribution is identical to a given distribution P, and whether the (joint) distribution is a product distribution. One way to arrive at the decision is to *learn the distribution* first – we saw that this entails collecting $\Theta(k/\delta^2)$ samples. In particular, the number of samples necessary is linear in the alphabet size k. However, for specific instances of property testing, there exist algorithms (and matching lower bounds) that work with *sublinear* (in k) number of samples.

4.4 Uniformity testing

We provide an algorithm that uses $O(\sqrt{k}/\delta^2)$ for uniformity testing, namely, testing whether the samples are drawn from a uniform distribution on an alphabet of size k or from a distribution that is δ-away from the uniform distribution in total variation distance. Denote by U_k the uniform distribution on $[k]$ and let

$$\overline{\mathcal{U}} := \{P : d_{\mathrm{TV}}(P, U_k) \geq \delta\}.$$

Formally, for $X_1, \ldots, X_n$ i.i.d. P, we seek to devise a test T that uses $O(\sqrt{k}/\delta^2)$ samples to distinguish between

$$\mathcal{H}_0 : P \equiv U_k$$
$$\mathcal{H}_1 : P \equiv Q$$

where $Q \in \overline{\mathcal{U}}$ such that

$$\max_{Q \in \overline{\mathcal{U}}} P_e(T | U_k, Q) \leq \frac{1}{3}.$$

Recall that it suffices to restrict ourselves to $1/3$ in the PAC framework by the *median trick* (see Example 2.3).

Next, we describe the test. The essential idea is that for $X_1, \ldots, X_n \sim$ i.i.d. U_k, if $n \ll \sqrt{k}$, then with high probability, the samples are all distinct. Further, the probability that two i.i.d. samples X and Y from a distribution P are equal,

$$\mathbb{P}[X = Y] = \|P\|_2^2,$$

referred to as the *collision probability* is minimized for the uniform distribution.

Exercise 4.9: Collision probability

1. Show that for X, Y, i.i.d. P,
$$\mathbb{P}[X = Y] = \|P\|_2^2.$$

2. Show that the collision probability is minimized for uniform distribution.
Hint: Show that
$$\|P\|_2^2 = \frac{1}{k} + \|P - U_k\|_2^2 \geq \frac{1}{k}.$$

The fact above suggests that we use the number of repeated values in $X_1, \ldots, X_n$ as the test statistic. For

$$Z = \frac{1}{\binom{n}{2}} \sum_{i<j} \mathbf{1}_{\{X_i = X_j\}},$$

we use the following test:

$$T(X^n) = \begin{cases} 0 & \text{if } Z \leq \tau \\ 1 & \text{otherwise.} \end{cases}$$

Note that the test statistic satisfies

$$\mathbb{E}Z = \sum_{i=1}^{k} P_i^2 = \|P\|_2^2.$$

Exercise 4.10: Distance from uniform

1. Show that
$$d_{\mathrm{TV}}(P, U_k) \leq \frac{\sqrt{k}}{2} \|P - U_k\|_2^2.$$
Hint: Cauchy-Schwarz inequality.

2. Using the inequality above, show that if $d_{\mathrm{TV}}(P, U_k) > \delta$, then
$$\|P - U_k\|_2^2 \geq \frac{4\delta^2}{k}.$$

3. Show that if $d_{\mathrm{TV}}(P, U_k) > \delta$, then
$$\|P\|_2^2 > \frac{1 + 4\delta^2}{k}.$$
Hint: Use Exercise 4.9.2.

For any distribution P, clearly, if $d_{\mathrm{TV}}(P, U_k) = 0$, then $\|P - U_k\|_2^2 = 0$ and by Exercise 4.9,

$$\|P\|_2^2 = \frac{1}{k}.$$

Further, from Exercise 4.10, if $d_{\mathrm{TV}}(P, U_k) > \delta$, then

$$\|P\|_2^2 > \frac{1 + 4\delta^2}{k}.$$

Hence, the hypothesis testing problem is essentially to distinguish between

$$\begin{aligned} \mathcal{H}_0 :& \|P\|_2^2 = \frac{1}{k} \\ \mathcal{H}_1 :& \|P\|_2^2 \geq \frac{1}{k} + \frac{4\delta^2}{k}. \end{aligned}$$

This prompts us to choose

$$\tau = \frac{1 + 2\delta^2}{k}.$$

Next, we control the variance of the test statistic Z.

$$\begin{aligned}
\mathrm{Var}(Z) &= \mathbb{E}Z^2 - (\mathbb{E}Z)^2 \\
&= \frac{1}{\binom{n}{2}^2}\left[6\binom{n}{4}\|P\|_2^4 + 6\binom{n}{3}\|P\|_3^3 + \binom{n}{2}\|P\|_2^2\right] - \|P\|_2^4 \\
&= \frac{1}{\binom{n}{2}^2}\left[\left\{6\binom{n}{4} - \binom{n}{2}^2\right\}\|P\|_2^4 + 6\binom{n}{3}\|P\|_3^3 + \binom{n}{2}\|P\|_2^2\right] \\
&\le \frac{4}{n}\|P\|_3^3 + \frac{4}{n^2}\|P\|_2^2 \\
&\le \frac{4}{n}(\mathbb{E}Z)^{3/2} + \frac{4}{n^2}\mathbb{E}Z
\end{aligned}$$

where the first inequality follows from

$$6\binom{n}{4} - \binom{n}{2}^2 < 0$$

and in the last step, we have used $\|P\|_3 \le \|P\|_2$. Under $\mathcal{H}_0$, $\mathbb{E}Z = 1/k$ and hence

$$\begin{aligned}
\mathbb{P}[Z \ge \tau] &= \mathbb{P}[Z \ge (1+2\delta^2)\mathbb{E}Z] \\
&\le \mathbb{P}[Z \ge (1+\delta^2)\mathbb{E}Z].
\end{aligned} \tag{4.24}$$

Under $\mathcal{H}_1$,

$$\begin{aligned}
\mathbb{P}[Z \le \tau] &= \mathbb{P}\left[Z \le \frac{(1-\delta^2)(1+2\delta^2)}{k}\right] \\
&\le \mathbb{P}[Z \le (1-\delta^2)\mathbb{E}Z]
\end{aligned} \tag{4.25}$$

where we have used

$$(1-\delta^2)(1+4\delta^2) \ge 1+2\delta^2 \qquad \text{for } \delta \le \frac{1}{2}$$

and

$$\mathbb{E}Z \ge \frac{1+4\delta^2}{k}. \qquad \text{(see Exercise 4.10.2)}$$

From (4.24) and (4.25), we have

$$\begin{aligned}
\mathbb{P}\left[|Z - \mathbb{E}Z| \ge \delta^2\mathbb{E}Z\right] &\le \frac{\mathrm{Var}(Z)}{\delta^4(\mathbb{E}Z)^2} \\
&\le \frac{4}{\delta^4 n(\mathbb{E}Z)^{1/2}} + \frac{4}{\delta^4 n^2 \mathbb{E}Z} \\
&\le \frac{4\sqrt{k}}{\delta^4 n} + \frac{4k}{\delta^4 n^2}
\end{aligned}$$

where the last inequality follows since under both the hypotheses, $\mathbb{E}Z \ge 1/k$. Thus, to ensure that the probability above is upper bounded by $1/3$, it suffices to have

$$n \ge \frac{c\sqrt{k}}{\delta^4}$$

where c is a constant. In order to obtain the *refined* bound of $\sqrt{k}/\delta^2$, we tighten the upper bound on $\mathrm{Var}(Z)$. We have, in fact

$$\mathrm{Var}(Z) \le \frac{4}{n}\left(\|P\|_3^3 - \|P\|_2^4\right) + \frac{4}{n^2}\|P\|_2^2.$$

For $P \equiv U_k$, we have $\|P\|_3^2 = \|P\|_2^4$ and hence

$$\mathrm{Var}(Z) \leq \frac{4}{n^2 k}.$$

Using this, we have

$$\begin{aligned}
\mathbb{P}[Z \geq \tau] &= \mathbb{P}[Z \geq (1 + 2\delta^2)\mathbb{E}Z] \\
&\leq \frac{\mathrm{Var}(Z)}{4\delta^2(\mathbb{E}Z)^2} \\
&\leq \frac{k}{\delta^4 n^2}
\end{aligned}$$

which yields

$$n \geq \frac{c\sqrt{k}}{\delta^2}.$$

The corresponding calculation for $Q \in \overline{\mathcal{U}}$ is slightly more tedious (and is omitted) but it essentially leads to the same bound whereby we conclude that the sample complexity of uniformity testing is $O(\sqrt{k}/\delta^2)$. The matching lower bound calculation is omitted.

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